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Technique for performing low-latency communication in wireless LAN system

Abstract

One embodiment according to the present specification relates to a method for performing low-latency communication. A receiving STA can receive information about an access type from a transmitting STA. The receiving STA can identify a transmittable access category on the basis of information about the access type. The receiving STA can transmit data about the transmittable access category to the transmitting STA.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2020/017556, filed on Dec. 3, 2020, which claims the benefit of earlier filing date and right of priority to Korean Application No. 10-2019-0160194, filed on Dec. 4, 2019, the contents of which are all hereby incorporated by reference herein in their entireties.

BACKGROUND

Field of the Disclosure

(2) This specification relates to a technique for transmitting and receiving data in wireless communication, and more particularly, to a method and apparatus for performing low-latency communication in a wireless LAN system.

Related Art

(3) A wireless local area network (WLAN) has been enhanced in various ways. For example, the IEEE 802.11ax standard has proposed an enhanced communication environment by using orthogonal frequency division multiple access (OFDMA) and downlink multi-user multiple input multiple output (DL MU MIMO) schemes.

(4) The present specification proposes a technical feature that can be utilized in a new communication standard. For example, the new communication standard may be an extreme high throughput (EHT) standard which is currently being discussed. The EHT standard may use an increased bandwidth, an enhanced PHY layer protocol data unit (PPDU) structure, an enhanced sequence, a hybrid automatic repeat request (HARQ) scheme, or the like, which is newly proposed. The EHT standard may be called the IEEE 802.11be standard.

SUMMARY

Technical Problem

(5) Recently, as wired/wireless traffic has exploded, time delay-sensitive traffic has also increased significantly. Among the time delay-sensitive traffic, real-time audio/video transmission accounts for a large proportion. According to the proliferation of multimedia devices, the need to support time delay-sensitive traffic even in a wireless environment has increased. However, in a wireless environment rather than a wired environment, since the transmission speed is lower than that of the wired environment and there is a problem of interference from the surroundings, various methods are required to support time delay-sensitive traffic.

(6) In particular, wireless LAN is a communication system that must compete equally in the Industrial Scientific and Medical (ISM) band without a channel monopoly by a central base station. Accordingly, it is relatively more difficult for a wireless LAN to support traffic sensitive to time delay, compared to other communications other than the wireless LAN. Accordingly, in the present specification, a technique for supporting traffic sensitive to time delay may be proposed.

Technical Solutions

(7) According to various embodiments, a method in a wireless local area network system may comprise receiving, from a transmitting STA, information related to an access type; identifying a transmittable access category, based on the information about the access type; and transmitting, to the transmitting STA, data related to the transmittable access category.

Technical Effects

(8) According to an embodiment of the present specification, a method for supporting traffic sensitive to time delay may be proposed. In order to receive time delay-sensitive traffic, the AP may limit the access categories that can be transmitted from the STA. The AP may transmit information about the access type to the STA. The STA may identify a transmittable access category based on the information about the access type, and transmit data regarding the transmittable access category.

(9) According to an embodiment of the present specification, since an access category transmittable by a STA is limited, network performance can be optimized, and time delay-sensitive traffic (or data) can be appropriately transmitted and received.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** shows an example of a transmitting apparatus and/or receiving apparatus of the present specification.
(2) FIG. **2** is a conceptual view illustrating the structure of a wireless local area network (WLAN).
(3) FIG. **3** illustrates a general link setup process.
(4) FIG. **4** illustrates an example of a PPDU used in an IEEE standard.
(5) FIG. **5** illustrates a layout of resource units (RUs) used in a band of 20 MHz.
(6) FIG. **6** illustrates a layout of RUs used in a band of 40 MHz.
(7) FIG. **7** illustrates a layout of RUs used in a band of 80 MHz.
(8) FIG. **8** illustrates a structure of an HE-SIG-B field.
(9) FIG. **9** illustrates an example in which a plurality of user STAs are allocated to the same RU through a MU-MIMO scheme.
(10) FIG. **10** illustrates an operation based on UL-MU.
(11) FIG. **11** illustrates an example of a trigger frame.
(12) FIG. **12** illustrates an example of a common information field of a trigger frame.
(13) FIG. **13** illustrates an example of a subfield included in a per user information field.
(14) FIG. **14** describes a technical feature of the UORA scheme.
(15) FIG. **15** illustrates an example of a channel used/supported/defined within a 2.4 GHz band.
(16) FIG. **16** illustrates an example of a channel used/supported/defined within a 5 GHz band.
(17) FIG. **17** illustrates an example of a channel used/supported/defined within a 6 GHz band.
(18) FIG. **18** illustrates an example of a PPDU used in the present specification.
(19) FIG. **19** illustrates an example of a modified transmission device and/or receiving device of the present specification.
(20) FIG. **20** is a flowchart related to an operation for performing low latency communication.
(21) FIG. **21** is a diagram illustrating four queues of EDCA.
(22) FIG. **22** is a flowchart illustrating an operation of a STA.
(23) FIG. **23** is a flowchart for explaining the operation of the AP.
(24) FIG. **24** is a flowchart illustrating an operation of a receiving STA.
(25) FIG. **25** is a flowchart illustrating an operation of a transmitting STA.
DETAILED DESCRIPTION
(26) In the present specification, “A or B” may mean “only A”, “only B” or “both A and B”. In other words, in the present specification, “A or B” may be interpreted as “A and/or B”. For example, in the present specification, “A, B, or C” may mean “only A”, “only B”, “only C”, or “any combination of A, B, C”.
(27) A slash (/) or comma used in the present specification may mean “and/or”. For example, “A/B” may mean “A and/or B”. Accordingly, “A/B” may mean “only A”, “only B”, or “both A and B”. For example, “A, B, C” may mean “A, B, or C”.
(28) In the present specification, “at least one of A and B” may mean “only A”, “only B”, or “both A and B”. In addition, in the present specification, the expression “at least one of A or B” or “at least one of A and/or B” may be interpreted as “at least one of A and B”.
(29) In addition, in the present specification, “at least one of A, B, and C” may mean “only A”, “only B”, “only C”, or “any combination of A, B, and C”. In addition, “at least one of A, B, or C” or “at least one of A, B, and/or C” may mean “at least one of A, B, and C”.
(30) In addition, a parenthesis used in the present specification may mean “for example”. Specifically, when indicated as “control information (EHT-signal)”, it may mean that “EHT-signal” is proposed as an example of the “control information”. In other words, the “control information” of the present specification is not limited to “EHT-signal”, and “EHT-signal” may be proposed as an example of the “control information”. In addition, when indicated as “control information (i.e., EHT-signal)”, it may also mean that “EHT-signal” is proposed as an example of the “control

information".
(31) Technical features described individually in one figure in the present specification may be individually implemented, or may be simultaneously implemented.
(32) The following example of the present specification may be applied to various wireless communication systems. For example, the following example of the present specification may be applied to a wireless local area network (WLAN) system. For example, the present specification may be applied to the IEEE 802.11a/g/n/ac standard or the IEEE 802.11ax standard. In addition, the present specification may also be applied to the newly proposed EHT standard or IEEE 802.11be standard. In addition, the example of the present specification may also be applied to a new WLAN standard enhanced from the EHT standard or the IEEE 802.11be standard. In addition, the example of the present specification may be applied to a mobile communication system. For example, it may be applied to a mobile communication system based on long term evolution (LTE) depending on a 3rd generation partnership project (3GPP) standard and based on evolution of the LTE. In addition, the example of the present specification may be applied to a communication system of a 5G NR standard based on the 3GPP standard.
(33) Hereinafter, in order to describe a technical feature of the present specification, a technical feature applicable to the present specification will be described.
(34) FIG. **1** shows an example of a transmitting apparatus and/or receiving apparatus of the present specification.
(35) In the example of FIG. **1**, various technical features described below may be performed. FIG. **1** relates to at least one station (STA). For example, STAs **110** and **120** of the present specification may also be called in various terms such as a mobile terminal, a wireless device, a wireless transmit/receive unit (WTRU), a user equipment (UE), a mobile station (MS), a mobile subscriber unit, or simply a user. The STAs **110** and **120** of the present specification may also be called in various terms such as a network, a base station, a node-B, an access point (AP), a repeater, a router, a relay, or the like. The STAs **110** and **120** of the present specification may also be referred to as various names such as a receiving apparatus, a transmitting apparatus, a receiving STA, a transmitting STA, a receiving device, a transmitting device, or the like.
(36) For example, the STAs **110** and **120** may serve as an AP or a non-AP. That is, the STAs **110** and **120** of the present specification may serve as the AP and/or the non-AP.
(37) STAs **110** and **120** of the present specification may support various communication standards together in addition to the IEEE 802.11 standard. For example, a communication standard (e.g., LTE, LTE-A, 5G NR standard) or the like based on the 3GPP standard may be supported. In addition, the STA of the present specification may be implemented as various devices such as a mobile phone, a vehicle, a personal computer, or the like. In addition, the STA of the present specification may support communication for various communication services such as voice calls, video calls, data communication, and self-driving (autonomous-driving), or the like.
(38) The STAs **110** and **120** of the present specification may include a medium access control (MAC) conforming to the IEEE 802.11 standard and a physical layer interface for a radio medium.
(39) The STAs **110** and **120** will be described below with reference to a sub-figure (a) of FIG. **1**.
(40) The first STA **110** may include a processor **111**, a memory **112**, and a transceiver **113**. The illustrated process, memory, and transceiver may be implemented individually as separate chips, or at least two blocks/functions may be implemented through a single chip.
(41) The transceiver **113** of the first STA performs a signal transmission/reception operation. Specifically, an IEEE 802.11 packet (e.g., IEEE 802.11a/b/g/n/ac/ax/be, etc.) may be transmitted/received.
(42) For example, the first STA **110** may perform an operation intended by an AP. For example, the processor **111** of the AP may receive a signal through the transceiver **113**, process a reception (RX) signal, generate a transmission (TX) signal, and provide control for signal transmission. The memory **112** of the AP may store a signal (e.g., RX signal) received through the transceiver **113**,

and may store a signal (e.g., TX signal) to be transmitted through the transceiver.

(43) For example, the second STA **120** may perform an operation intended by a non-AP STA. For example, a transceiver **123** of a non-AP performs a signal transmission/reception operation. Specifically, an IEEE 802.11 packet (e.g., IEEE 802.11a/b/g/n/ac/ax/be packet, etc.) may be transmitted/received.

(44) For example, a processor **121** of the non-AP STA may receive a signal through the transceiver **123**, process an RX signal, generate a TX signal, and provide control for signal transmission. A memory **122** of the non-AP STA may store a signal (e.g., RX signal) received through the transceiver **123**, and may store a signal (e.g., TX signal) to be transmitted through the transceiver.

(45) For example, an operation of a device indicated as an AP in the specification described below may be performed in the first STA **110** or the second STA **120**. For example, if the first STA **110** is the AP, the operation of the device indicated as the AP may be controlled by the processor **111** of the first STA **110**, and a related signal may be transmitted or received through the transceiver **113** controlled by the processor **111** of the first STA **110**. In addition, control information related to the operation of the AP or a TX/RX signal of the AP may be stored in the memory **112** of the first STA **110**. In addition, if the second STA **120** is the AP, the operation of the device indicated as the AP may be controlled by the processor **121** of the second STA **120**, and a related signal may be transmitted or received through the transceiver **123** controlled by the processor **121** of the second STA **120**. In addition, control information related to the operation of the AP or a TX/RX signal of the AP may be stored in the memory **122** of the second STA **120**.

(46) For example, in the specification described below, an operation of a device indicated as a non-AP (or user-STA) may be performed in the first STA **110** or the second STA **120**. For example, if the second STA **120** is the non-AP, the operation of the device indicated as the non-AP may be controlled by the processor **121** of the second STA **120**, and a related signal may be transmitted or received through the transceiver **123** controlled by the processor **121** of the second STA **120**. In addition, control information related to the operation of the non-AP or a TX/RX signal of the non-AP may be stored in the memory **122** of the second STA **120**. For example, if the first STA **110** is the non-AP, the operation of the device indicated as the non-AP may be controlled by the processor **111** of the first STA **110**, and a related signal may be transmitted or received through the transceiver **113** controlled by the processor **111** of the first STA **110**. In addition, control information related to the operation of the non-AP or a TX/RX signal of the non-AP may be stored in the memory **112** of the first STA **110**.

(47) In the specification described below, a device called a (transmitting/receiving) STA, a first STA, a second STA, a STA1, a STA2, an AP, a first AP, a second AP, an AP1, an AP2, a (transmitting/receiving) terminal, a (transmitting/receiving) device, a (transmitting/receiving) apparatus, a network, or the like may imply the STAs **110** and **120** of FIG. **1**. For example, a device indicated as, without a specific reference numeral, the (transmitting/receiving) STA, the first STA, the second STA, the STA1, the STA2, the AP, the first AP, the second AP, the AP1, the AP2, the (transmitting/receiving) terminal, the (transmitting/receiving) device, the (transmitting/receiving) apparatus, the network, or the like may imply the STAs **110** and **120** of FIG. **1**. For example, in the following example, an operation in which various STAs transmit/receive a signal (e.g., a PPDU) may be performed in the transceivers **113** and **123** of FIG. **1**. In addition, in the following example, an operation in which various STAs generate a TX/RX signal or perform data processing and computation in advance for the TX/RX signal may be performed in the processors **111** and **121** of FIG. **1**. For example, an example of an operation for generating the TX/RX signal or performing the data processing and computation in advance may include: 1) an operation of determining/obtaining/configuring/computing/decoding/encoding bit information of a sub-field (SIG, STF, LTF, Data) included in a PPDU; 2) an operation of determining/configuring/obtaining a time resource or frequency resource (e.g., a subcarrier resource) or the like used for the sub-field (SIG, STF, LTF, Data) included the PPDU; 3) an operation of determining/configuring/obtaining a

specific sequence (e.g., a pilot sequence, an STF/LTF sequence, an extra sequence applied to SIG) or the like used for the sub-field (SIG, STF, LTF, Data) field included in the PPDU; 4) a power control operation and/or power saving operation applied for the STA; and 5) an operation related to determining/obtaining/configuring/decoding/encoding or the like of an ACK signal. In addition, in the following example, a variety of information used by various STAs for determining/obtaining/configuring/computing/decoding/decoding a TX/RX signal (e.g., information related to a field/subfield/control field/parameter/power or the like) may be stored in the memories **112** and **122** of FIG. **1**.

(48) The aforementioned device/STA of the sub-figure (a) of FIG. **1** may be modified as shown in the sub-figure (b) of FIG. **1**. Hereinafter, the STAs **110** and **120** of the present specification will be described based on the sub-figure (b) of FIG. **1**.

(49) For example, the transceivers **113** and **123** illustrated in the sub-figure (b) of FIG. **1** may perform the same function as the aforementioned transceiver illustrated in the sub-figure (a) of FIG. **1**. For example, processing chips **114** and **124** illustrated in the sub-figure (b) of FIG. **1** may include the processors **111** and **121** and the memories **112** and **122**. The processors **111** and **121** and memories **112** and **122** illustrated in the sub-figure (b) of FIG. **1** may perform the same function as the aforementioned processors **111** and **121** and memories **112** and **122** illustrated in the sub-figure (a) of FIG. **1**.

(50) A mobile terminal, a wireless device, a wireless transmit/receive unit (WTRU), a user equipment (UE), a mobile station (MS), a mobile subscriber unit, a user, a user STA, a network, a base station, a Node-B, an access point (AP), a repeater, a router, a relay, a receiving unit, a transmitting unit, a receiving STA, a transmitting STA, a receiving device, a transmitting device, a receiving apparatus, and/or a transmitting apparatus, which are described below, may imply the STAs **110** and **120** illustrated in the sub-figure (a)/(b) of FIG. **1**, or may imply the processing chips **114** and **124** illustrated in the sub-figure (b) of FIG. **1**. That is, a technical feature of the present specification may be performed in the STAs **110** and **120** illustrated in the sub-figure (a)/(b) of FIG. **1**, or may be performed only in the processing chips **114** and **124** illustrated in the sub-figure (b) of FIG. **1**. For example, a technical feature in which the transmitting STA transmits a control signal may be understood as a technical feature in which a control signal generated in the processors **111** and **121** illustrated in the sub-figure (a)/(b) of FIG. **1** is transmitted through the transceivers **113** and **123** illustrated in the sub-figure (a)/(b) of FIG. **1**. Alternatively, the technical feature in which the transmitting STA transmits the control signal may be understood as a technical feature in which the control signal to be transferred to the transceivers **113** and **123** is generated in the processing chips **114** and **124** illustrated in the sub-figure (b) of FIG. **1**.

(51) For example, a technical feature in which the receiving STA receives the control signal may be understood as a technical feature in which the control signal is received by means of the transceivers **113** and **123** illustrated in the sub-figure (a) of FIG. **1**. Alternatively, the technical feature in which the receiving STA receives the control signal may be understood as the technical feature in which the control signal received in the transceivers **113** and **123** illustrated in the sub-figure (a) of FIG. **1** is obtained by the processors **111** and **121** illustrated in the sub-figure (a) of FIG. **1**. Alternatively, the technical feature in which the receiving STA receives the control signal may be understood as the technical feature in which the control signal received in the transceivers **113** and **123** illustrated in the sub-figure (b) of FIG. **1** is obtained by the processing chips **114** and **124** illustrated in the sub-figure (b) of FIG. **1**.

(52) Referring to the sub-figure (b) of FIG. **1**, software codes **115** and **125** may be included in the memories **112** and **122**. The software codes **115** and **126** may include instructions for controlling an operation of the processors **111** and **121**. The software codes **115** and **125** may be included as various programming languages.

(53) The processors **111** and **121** or processing chips **114** and **124** of FIG. **1** may include an application-specific integrated circuit (ASIC), other chipsets, a logic circuit and/or a data

processing device. The processor may be an application processor (AP). For example, the processors **111** and **121** or processing chips **114** and **124** of FIG. **1** may include at least one of a digital signal processor (DSP), a central processing unit (CPU), a graphics processing unit (GPU), and a modulator and demodulator (modem). For example, the processors **111** and **121** or processing chips **114** and **124** of FIG. **1** may be SNAPDRAGON™ series of processors made by Qualcomm®, EXYNOS™ series of processors made by Samsung®, A series of processors made by Apple®, HELIO™ series of processors made by MediaTek®, ATOM™ series of processors made by Intel® or processors enhanced from these processors.

(54) In the present specification, an uplink may imply a link for communication from a non-AP STA to an SP STA, and an uplink PPDU/packet/signal or the like may be transmitted through the uplink. In addition, in the present specification, a downlink may imply a link for communication from the AP STA to the non-AP STA, and a downlink PPDU/packet/signal or the like may be transmitted through the downlink.

(55) FIG. **2** is a conceptual view illustrating the structure of a wireless local area network (WLAN).

(56) An upper part of FIG. **2** illustrates the structure of an infrastructure basic service set (BSS) of institute of electrical and electronic engineers (i.e. EE) 802.11.

(57) Referring the upper part of FIG. **2**, the wireless LAN system may include one or more infrastructure BSSs **200** and **205** (hereinafter, referred to as BSS). The BSSs **200** and **205** as a set of an AP and a STA such as an access point (AP) **225** and a station (STA1) **200-1** which are successfully synchronized to communicate with each other are not concepts indicating a specific region. The BSS **205** may include one or more STAs **205-1** and **205-2** which may be joined to one AP **230**.

(58) The BSS may include at least one STA, APs providing a distribution service, and a distribution system (DS) **210** connecting multiple APs.

(59) The distribution system **210** may implement an extended service set (ESS) **240** extended by connecting the multiple BSSs **200** and **205**. The ESS **240** may be used as a term indicating one network configured by connecting one or more APs **225** or **230** through the distribution system **210**. The AP included in one ESS **240** may have the same service set identification (SSID).

(60) A portal **220** may serve as a bridge which connects the wireless LAN network (i.e. EE 802.11) and another network (e.g., 802.X).

(61) In the BSS illustrated in the upper part of FIG. **2**, a network between the APs **225** and **230** and a network between the APs **225** and **230** and the STAs **200-1**, **205-1**, and **205-2** may be implemented. However, the network is configured even between the STAs without the APs **225** and **230** to perform communication. A network in which the communication is performed by configuring the network even between the STAs without the APs **225** and **230** is defined as an Ad-Hoc network or an independent basic service set (IBSS).

(62) A lower part of FIG. **2** illustrates a conceptual view illustrating the IBSS.

(63) Referring to the lower part of FIG. **2**, the IBSS is a BSS that operates in an Ad-Hoc mode. Since the IBSS does not include the access point (AP), a centralized management entity that performs a management function at the center does not exist. That is, in the IBSS, STAs **250-1**, **250-2**, **250-3**, **255-4**, and **255-5** are managed by a distributed manner. In the IBSS, all STAs **250-1**, **250-2**, **250-3**, **255-4**, and **255-5** may be constituted by movable STAs and are not permitted to access the DS to constitute a self-contained network.

(64) FIG. **3** illustrates a general link setup process.

(65) In S**310**, a STA may perform a network discovery operation. The network discovery operation may include a scanning operation of the STA. That is, to access a network, the STA needs to discover a participating network. The STA needs to identify a compatible network before participating in a wireless network, and a process of identifying a network present in a particular area is referred to as scanning. Scanning methods include active scanning and passive scanning.

(66) FIG. **3** illustrates a network discovery operation including an active scanning process. In

active scanning, a STA performing scanning transmits a probe request frame and waits for a response to the probe request frame in order to identify which AP is present around while moving to channels. A responder transmits a probe response frame as a response to the probe request frame to the STA having transmitted the probe request frame. Here, the responder may be a STA that transmits the last beacon frame in a BSS of a channel being scanned. In the BSS, since an AP transmits a beacon frame, the AP is the responder. In an IBSS, since STAs in the IBSS transmit a beacon frame in turns, the responder is not fixed. For example, when the STA transmits a probe request frame via channel 1 and receives a probe response frame via channel 1, the STA may store BSS-related information included in the received probe response frame, may move to the next channel (e.g., channel 2), and may perform scanning (e.g., transmits a probe request and receives a probe response via channel 2) by the same method.
(67) Although not shown in FIG. **3**, scanning may be performed by a passive scanning method. In passive scanning, a STA performing scanning may wait for a beacon frame while moving to channels. A beacon frame is one of management frames in IEEE 802.11 and is periodically transmitted to indicate the presence of a wireless network and to enable the STA performing scanning to find the wireless network and to participate in the wireless network. In a BSS, an AP serves to periodically transmit a beacon frame. In an IBSS, STAs in the IBSS transmit a beacon frame in turns. Upon receiving the beacon frame, the STA performing scanning stores information about a BSS included in the beacon frame and records beacon frame information in each channel while moving to another channel. The STA having received the beacon frame may store BSS-related information included in the received beacon frame, may move to the next channel, and may perform scanning in the next channel by the same method.
(68) After discovering the network, the STA may perform an authentication process in S**320**. The authentication process may be referred to as a first authentication process to be clearly distinguished from the following security setup operation in S**340**. The authentication process in S**320** may include a process in which the STA transmits an authentication request frame to the AP and the AP transmits an authentication response frame to the STA in response. The authentication frames used for an authentication request/response are management frames.
(69) The authentication frames may include information about an authentication algorithm number, an authentication transaction sequence number, a status code, a challenge text, a robust security network (RSN), and a finite cyclic group.
(70) The STA may transmit the authentication request frame to the AP. The AP may determine whether to allow the authentication of the STA based on the information included in the received authentication request frame. The AP may provide the authentication processing result to the STA via the authentication response frame.
(71) When the STA is successfully authenticated, the STA may perform an association process in S**330**. The association process includes a process in which the STA transmits an association request frame to the AP and the AP transmits an association response frame to the STA in response. The association request frame may include, for example, information about various capabilities, a beacon listen interval, a service set identifier (SSID), a supported rate, a supported channel, RSN, a mobility domain, a supported operating class, a traffic indication map (TIM) broadcast request, and an interworking service capability. The association response frame may include, for example, information about various capabilities, a status code, an association ID (AID), a supported rate, an enhanced distributed channel access (EDCA) parameter set, a received channel power indicator (RCPI), a received signal-to-noise indicator (RSNI), a mobility domain, a timeout interval (association comeback time), an overlapping BSS scanning parameter, a TIM broadcast response, and a QoS map.
(72) In S**340**, the STA may perform a security setup process. The security setup process in S**340** may include a process of setting up a private key through four-way handshaking, for example, through an extensible authentication protocol over LAN (EAPOL) frame.

(73) FIG. **4** illustrates an example of a PPDU used in an IEEE standard.
(74) As illustrated, various types of PHY protocol data units (PPDUs) are used in IEEE a/g/n/ac standards. Specifically, an LTF and a STF include a training signal, a SIG-A and a SIG-B include control information for a receiving STA, and a data field includes user data corresponding to a PSDU (MAC PDU/aggregated MAC PDU).
(75) FIG. **4** also includes an example of an HE PPDU according to IEEE 802.11ax. The HE PPDU according to FIG. **4** is an illustrative PPDU for multiple users. An HE-SIG-B may be included only in a PPDU for multiple users, and an HE-SIG-B may be omitted in a PPDU for a single user.
(76) As illustrated in FIG. **4**, the HE-PPDU for multiple users (MUs) may include a legacy-short training field (L-STF), a legacy-long training field (L-LTF), a legacy-signal (L-SIG), a high efficiency-signal A (HE-SIG A), a high efficiency-signal-B (HE-SIG B), a high efficiency-short training field (HE-STF), a high efficiency-long training field (HE-LTF), a data field (alternatively, an MAC payload), and a packet extension (PE) field. The respective fields may be transmitted for illustrated time periods (i.e., 4 or 8 μs).
(77) Hereinafter, a resource unit (RU) used for a PPDU is described. An RU may include a plurality of subcarriers (or tones). An RU may be used to transmit a signal to a plurality of STAs according to OFDMA. Further, an RU may also be defined to transmit a signal to one STA. An RU may be used for an STF, an LTF, a data field, or the like.
(78) FIG. **5** illustrates a layout of resource units (RUs) used in a band of 20 MHz.
(79) As illustrated in FIG. **5**, resource units (RUs) corresponding to different numbers of tones (i.e., subcarriers) may be used to form some fields of an HE-PPDU. For example, resources may be allocated in illustrated RUs for an HE-STF, an HE-LTF, and a data field.
(80) As illustrated in the uppermost part of FIG. **5**, a 26-unit (i.e., a unit corresponding to 26 tones) may be disposed. Six tones may be used for a guard band in the leftmost band of the 20 MHz band, and five tones may be used for a guard band in the rightmost band of the 20 MHz band. Further, seven DC tones may be inserted in a center band, that is, a DC band, and a 26-unit corresponding to 13 tones on each of the left and right sides of the DC band may be disposed. A 26-unit, a 52-unit, and a 106-unit may be allocated to other bands. Each unit may be allocated for a receiving STA, that is, a user.
(81) The layout of the RUs in FIG. **5** may be used not only for a multiple users (MUs) but also for a single user (SU), in which case one 242-unit may be used and three DC tones may be inserted as illustrated in the lowermost part of FIG. **5**.
(82) Although FIG. **5** proposes RUs having various sizes, that is, a 26-RU, a 52-RU, a 106-RU, and a 242-RU, specific sizes of RUs may be extended or increased. Therefore, the present embodiment is not limited to the specific size of each RU (i.e., the number of corresponding tones).
(83) FIG. **6** illustrates a layout of RUs used in a band of 40 MHz.
(84) Similarly to FIG. **5** in which RUs having various sizes are used, a 26-RU, a 52-RU, a 106-RU, a 242-RU, a 484-RU, and the like may be used in an example of FIG. **6**. Further, five DC tones may be inserted in a center frequency, 12 tones may be used for a guard band in the leftmost band of the 40 MHz band, and 11 tones may be used for a guard band in the rightmost band of the 40 MHz band.
(85) As illustrated in FIG. **6**, when the layout of the RUs is used for a single user, a 484-RU may be used. The specific number of RUs may be changed similarly to FIG. **5**.
(86) FIG. **7** illustrates a layout of RUs used in a band of 80 MHz.
(87) Similarly to FIG. **5** and FIG. **6** in which RUs having various sizes are used, a 26-RU, a 52-RU, a 106-RU, a 242-RU, a 484-RU, a 996-RU, and the like may be used in an example of FIG. **7**. Further, seven DC tones may be inserted in the center frequency, 12 tones may be used for a guard band in the leftmost band of the 80 MHz band, and 11 tones may be used for a guard band in the rightmost band of the 80 MHz band. In addition, a 26-RU corresponding to 13 tones on each of the left and right sides of the DC band may be used.

(88) As illustrated in FIG. **7**, when the layout of the RUs is used for a single user, a 996-RU may be used, in which case five DC tones may be inserted.
(89) In the meantime, the fact that the specific number of RUs can be changed is the same as those of FIGS. **5** and **6**.
(90) The RU arrangement (i.e., RU location) shown in FIGS. **5** to **7** can be applied to a new wireless LAN system (e.g. EHT system) as it is. Meanwhile, for the 160 MHz band supported by the new WLAN system, the RU arrangement for 80 MHz (i.e., an example of FIG. **7**) may be repeated twice, or the RU arrangement for the 40 MHz (i.e., an example of FIG. **6**) may be repeated 4 times. In addition, when the EHT PPDU is configured for the 320 MHz band, the arrangement of the RU for 80 MHz (i.e., an example of FIG. **7**) may be repeated 4 times or the arrangement of the RU for 40 MHz (i.e., an example of FIG. **6**) may be repeated 8 times.
(91) One RU of the present specification may be allocated for a single STA (e.g., a single non-AP STA). Alternatively, a plurality of RUs may be allocated for one STA (e.g., a non-AP STA).
(92) The RU described in the present specification may be used in uplink (UL) communication and downlink (DL) communication. For example, when UL-MU communication which is solicited by a trigger frame is performed, a transmitting STA (e.g., an AP) may allocate a first RU (e.g., 26/52/106/242-RU, etc.) to a first STA through the trigger frame, and may allocate a second RU (e.g., 26/52/106/242-RU, etc.) to a second STA. Thereafter, the first STA may transmit a first trigger-based PPDU based on the first RU, and the second STA may transmit a second trigger-based PPDU based on the second RU. The first/second trigger-based PPDU is transmitted to the AP at the same (or overlapped) time period.
(93) For example, when a DL MU PPDU is configured, the transmitting STA (e.g., AP) may allocate the first RU (e.g., 26/52/106/242-RU. etc.) to the first STA, and may allocate the second RU (e.g., 26/52/106/242-RU, etc.) to the second STA. That is, the transmitting STA (e.g., AP) may transmit HE-STF, HE-LTF, and Data fields for the first STA through the first RU in one MU PPDU, and may transmit HE-STF, HE-LTF, and Data fields for the second STA through the second RU.
(94) Information related to a layout of the RU may be signaled through HE-SIG-B.
(95) FIG. **8** illustrates a structure of an HE-SIG-B field.
(96) As illustrated, an HE-SIG-B field **810** includes a common field **820** and a user-specific field **830**. The common field **820** may include information commonly applied to all users (i.e., user STAs) which receive SIG-B. The user-specific field **830** may be called a user-specific control field. When the SIG-B is transferred to a plurality of users, the user-specific field **830** may be applied only any one of the plurality of users.
(97) As illustrated in FIG. **8**, the common field **820** and the user-specific field **830** may be separately encoded.
(98) The common field **820** may include RU allocation information of N*8 bits. For example, the RU allocation information may include information related to a location of an RU. For example, when a 20 MHz channel is used as shown in FIG. **5**, the RU allocation information may include information related to a specific frequency band to which a specific RU (26-RU/52-RU/106-RU) is arranged.
(99) An example of a case in which the RU allocation information consists of 8 bits is as follows.
(100) TABLE-US-00001 TABLE 1 8 bits indices (B7 B6 B5 B4 Number B3 B2 B1 B0) #1 #2 #3 #4 #5 #6 #7 #8 #9 of entries 00000000 26 26 26 26 26 26 26 26 26 1 00000001 26 26 26 26 26 26 26 52 1 00000010 26 26 26 26 26 52 26 26 1 00000011 26 26 26 26 26 52 52 1 00000100 26 26 52 26 26 26 26 26 1 00000101 26 26 52 26 26 26 52 1 00000110 26 26 52 26 52 26 26 1 00000111 26 26 52 26 52 52 1 00001000 52 26 26 26 26 26 26 26 1
(101) As shown the example of FIG. **5**, up to nine 26-RUs may be allocated to the 20 MHz channel. When the RU allocation information of the common field **820** is set to “00000000” as shown in Table 1, the nine 26-RUs may be allocated to a corresponding channel (i.e., 20 MHz). In addition, when the RU allocation information of the common field **820** is set to “00000001” as

shown in Table 1, seven 26-RUs and one 52-RU are arranged in a corresponding channel. That is, in the example of FIG. **5**, the 52-RU may be allocated to the rightmost side, and the seven 26-RUs may be allocated to the left thereof.
(102) The example of Table 1 shows only some of RU locations capable of displaying the RU allocation information.
(103) For example, the RU allocation information may include an example of Table 2 below.
(104) TABLE-US-00002 TABLE 2 8 bits indices (B7 B6 B5 B4 Number B3 B2 B1 B0) #1 #2 #3 #4 #5 #6 #7 #8 #9 of entries 01000y.sub.2y.sub.1y.sub.0 106 26 26 26 26 26 8 01001y.sub.2y.sub.1y.sub.0 106 26 26 26 52 8
(105) "01000y2y1y0" relates to an example in which a 106-RU is allocated to the leftmost side of the 20 MHz channel, and five 26-RUs are allocated to the right side thereof. In this case, a plurality of STAs (e.g., user-STAs) may be allocated to the 106-RU, based on a MU-MIMO scheme. Specifically, up to 8 STAs (e.g., user-STAs) may be allocated to the 106-RU, and the number of STAs (e.g., user-STAs) allocated to the 106-RU is determined based on 3-bit information (y2y1y0). For example, when the 3-bit information (y2y1y0) is set to N, the number of STAs (e.g., user-STAs) allocated to the 106-RU based on the MU-MIMO scheme may be N+1.
(106) In general, a plurality of STAs (e.g., user STAs) different from each other may be allocated to a plurality of RUs. However, the plurality of STAs (e.g., user STAs) may be allocated to one or more RUs having at least a specific size (e.g., 106 subcarriers), based on the MU-MIMO scheme.
(107) As shown in FIG. **8**, the user-specific field **830** may include a plurality of user fields. As described above, the number of STAs (e.g., user STAs) allocated to a specific channel may be determined based on the RU allocation information of the common field **820**. For example, when the RU allocation information of the common field **820** is "00000000", one user STA may be allocated to each of nine 26-RUs (e.g., nine user STAs may be allocated). That is, up to 9 user STAs may be allocated to a specific channel through an OFDMA scheme. In other words, up to 9 user STAs may be allocated to a specific channel through a non-MU-MIMO scheme.
(108) For example, when RU allocation is set to "01000y2y1y0", a plurality of STAs may be allocated to the 106-RU arranged at the leftmost side through the MU-MIMO scheme, and five user STAs may be allocated to five 26-RUs arranged to the right side thereof through the non-MU MIMO scheme. This case is specified through an example of FIG. **9**.
(109) FIG. **9** illustrates an example in which a plurality of user STAs are allocated to the same RU through a MU-MIMO scheme.
(110) For example, when RU allocation is set to "01000010" as shown in FIG. **9**, a 106-RU may be allocated to the leftmost side of a specific channel, and five 26-RUs may be allocated to the right side thereof. In addition, three user STAs may be allocated to the 106-RU through the MU-MIMO scheme. As a result, since eight user STAs are allocated, the user-specific field **830** of HE-SIG-B may include eight user fields.
(111) The eight user fields may be expressed in the order shown in FIG. **9**. In addition, as shown in FIG. **8**, two user fields may be implemented with one user block field.
(112) The user fields shown in FIG. **8** and FIG. **9** may be configured based on two formats. That is, a user field related to a MU-MIMO scheme may be configured in a first format, and a user field related to a non-MIMO scheme may be configured in a second format. Referring to the example of FIG. **9**, a user field 1 to a user field 3 may be based on the first format, and a user field 4 to a user field 8 may be based on the second format. The first format or the second format may include bit information of the same length (e.g., 21 bits).
(113) Each user field may have the same size (e.g., 21 bits). For example, the user field of the first format (the first of the MU-MIMO scheme) may be configured as follows.
(114) For example, a first bit (i.e., B0-B10) in the user field (i.e., 21 bits) may include identification information (e.g., STA-ID, partial AID, etc.) of a user STA to which a corresponding user field is allocated. In addition, a second bit (i.e., B11-B14) in the user field (i.e., 21 bits) may include

information related to a spatial configuration. Specifically, an example of the second bit (i.e., B11-B14) may be as shown in Table 3 and Table 4 below.
(115) TABLE-US-00003 TABLE 3 N.sub.STS N.sub.STS N.sub.STS N.sub.STS N.sub.STS N.sub.STS N.sub.STS N.sub.STS Total Number N.sub.user B3 . . . B0 [1] [2] [3] [4] [5] [6] [7] [8] N.sub.STS of entries 2 0000-0011 1-4 1 2-5 10 0100-0110 2-4 2 4-6 0111-1000 3-4 3 6-7 1001 4 4 8 3 0000-0011 1-4 1 1 3-6 13 0100-0110 2-4 2 1 5-7 0111-1000 3-4 3 1 7-8 1001-1011 2-4 2 2 6-8 1100 3 3 2 8 4 0000-0011 1-4 1 1 1 4-7 11 0100-0110 2-4 2 1 1 6-8 0111 3 3 1 1 8 1000-1001 2-3 2 2 1 7-8 1010 2 2 2 2 8
(116) TABLE-US-00004 TABLE 4 N.sub.STS N.sub.STS N.sub.STS N.sub.STS N.sub.STS N.sub.STS N.sub.STS N.sub.STS Total Number N.sub.user B3 . . . B0 [1] [2] [3] [4] [5] [6] [7] [8] N.sub.STS of entries 5 0000-0011 1-4 1 1 1 1 5-8 7 0100-0101 2-3 2 1 1 1 7-8 0110 2 2 2 1 1 8 6 0000-0010 1-3 1 1 1 1 1 6-8 4 0011 2 2 1 1 1 1 8 7 0000-0001 1-2 1 1 1 1 1 1 7-8 2 8 0000 1 1 1 1 1 1 1 1 8 1
(117) As shown in Table 3 and/or Table 4, the second bit (e.g., B11-B14) may include information related to the number of spatial streams allocated to the plurality of user STAs which are allocated based on the MU-MIMO scheme. For example, when three user STAs are allocated to the 106-RU based on the MU-MIMO scheme as shown in FIG. **9**, N_user is set to “3”. Therefore, values of N_STS[1], N_STS[2], and N_STS[3] may be determined as shown in Table 3. For example, when a value of the second bit (B11-B14) is “0011”, it may be set to N_STS[1]=4, N_STS[2]=1, N_STS[3]=1. That is, in the example of FIG. **9**, four spatial streams may be allocated to the user field 1, one spatial stream may be allocated to the user field 1, and one spatial stream may be allocated to the user field 3.
(118) As shown in the example of Table 3 and/or Table 4, information (i.e., the second bit, B11-B14) related to the number of spatial streams for the user STA may consist of 4 bits. In addition, the information (i.e., the second bit, B11-B14) on the number of spatial streams for the user STA may support up to eight spatial streams. In addition, the information (i.e., the second bit, B11-B14) on the number of spatial streams for the user STA may support up to four spatial streams for one user STA.
(119) In addition, a third bit (i.e., B15-18) in the user field (i.e., 21 bits) may include modulation and coding scheme (MCS) information. The MCS information may be applied to a data field in a PPDU including corresponding SIG-B.
(120) An MCS, MCS information, an MCS index, an MCS field, or the like used in the present specification may be indicated by an index value. For example, the MCS information may be indicated by an index 0 to an index 11. The MCS information may include information related to a constellation modulation type (e.g., BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM, 1024-QAM, etc.) and information related to a coding rate (e.g., 1/2, 2/3, 3/4, 5/6e, etc.). Information related to a channel coding type (e.g., LCC or LDPC) may be excluded in the MCS information.
(121) In addition, a fourth bit (i.e., B19) in the user field (i.e., 21 bits) may be a reserved field.
(122) In addition, a fifth bit (i.e., B20) in the user field (i.e., 21 bits) may include information related to a coding type (e.g., BCC or LDPC). That is, the fifth bit (i.e., B20) may include information related to a type (e.g., BCC or LDPC) of channel coding applied to the data field in the PPDU including the corresponding SIG-B.
(123) The aforementioned example relates to the user field of the first format (the format of the MU-MIMO scheme). An example of the user field of the second format (the format of the non-MU-MIMO scheme) is as follows.
(124) A first bit (e.g., B0-B10) in the user field of the second format may include identification information of a user STA. In addition, a second bit (e.g., B11-B13) in the user field of the second format may include information related to the number of spatial streams applied to a corresponding RU. In addition, a third bit (e.g., B14) in the user field of the second format may include information related to whether a beamforming steering matrix is applied. A fourth bit (e.g., B15-

B18) in the user field of the second format may include modulation and coding scheme (MCS) information. In addition, a fifth bit (e.g., B19) in the user field of the second format may include information related to whether dual carrier modulation (DCM) is applied. In addition, a sixth bit (i.e., B20) in the user field of the second format may include information related to a coding type (e.g., BCC or LDPC).

(125) FIG. **10** illustrates an operation based on UL-MU. As illustrated, a transmitting STA (e.g., an AP) may perform channel access through contending (e.g., a backoff operation), and may transmit a trigger frame **1030**. That is, the transmitting STA may transmit a PPDU including the trigger frame **1030**. Upon receiving the PPDU including the trigger frame, a trigger-based (TB) PPDU is transmitted after a delay corresponding to SIFS.

(126) TB PPDUs **1041** and **1042** may be transmitted at the same time period, and may be transmitted from a plurality of STAs (e.g., user STAs) having AIDs indicated in the trigger frame **1030**. An ACK frame **1050** for the TB PPDU may be implemented in various forms.

(127) A specific feature of the trigger frame is described with reference to FIG. **11** to FIG. **13**. Even if UL-MU communication is used, an orthogonal frequency division multiple access (OFDMA) scheme or a MU MIMO scheme may be used, and the OFDMA and MU-MIMO schemes may be simultaneously used.

(128) FIG. **11** illustrates an example of a trigger frame. The trigger frame of FIG. **11** allocates a resource for uplink multiple-user (MU) transmission, and may be transmitted, for example, from an AP. The trigger frame may be configured of a MAC frame, and may be included in a PPDU.

(129) Each field shown in FIG. **11** may be partially omitted, and another field may be added. In addition, a length of each field may be changed to be different from that shown in the figure.

(130) A frame control field **1110** of FIG. **11** may include information related to a MAC protocol version and extra additional control information. A duration field **1120** may include time information for NAV configuration or information related to an identifier (e.g., AID) of a STA.

(131) In addition, an RA field **1130** may include address information of a receiving STA of a corresponding trigger frame, and may be optionally omitted. A TA field **1140** may include address information of a STA (e.g., an AP) which transmits the corresponding trigger frame. A common information field **1150** includes common control information applied to the receiving STA which receives the corresponding trigger frame. For example, a field indicating a length of an L-SIG field of an uplink PPDU transmitted in response to the corresponding trigger frame or information for controlling content of a SIG-A field (i.e., HE-SIG-A field) of the uplink PPDU transmitted in response to the corresponding trigger frame may be included. In addition, as common control information, information related to a length of a CP of the uplink PPDU transmitted in response to the corresponding trigger frame or information related to a length of an LTF field may be included.

(132) In addition, per user information fields **1160#1** to **1160**#N corresponding to the number of receiving STAs which receive the trigger frame of FIG. **11** are preferably included. The per user information field may also be called an “allocation field”.

(133) In addition, the trigger frame of FIG. **11** may include a padding field **1170** and a frame check sequence field **1180**.

(134) Each of the per user information fields **1160#1** to **1160**#N shown in FIG. **11** may include a plurality of subfields.

(135) FIG. **12** illustrates an example of a common information field of a trigger frame. A subfield of FIG. **12** may be partially omitted, and an extra subfield may be added. In addition, a length of each subfield illustrated may be changed.

(136) A length field **1210** illustrated has the same value as a length field of an L-SIG field of an uplink PPDU transmitted in response to a corresponding trigger frame, and a length field of the L-SIG field of the uplink PPDU indicates a length of the uplink PPDU. As a result, the length field **1210** of the trigger frame may be used to indicate the length of the corresponding uplink PPDU.

(137) In addition, a cascade identifier field **1220** indicates whether a cascade operation is

performed. The cascade operation implies that downlink MU transmission and uplink MU transmission are performed together in the same TXOP. That is, it implies that downlink MU transmission is performed and thereafter uplink MU transmission is performed after a pre-set time (e.g., SIFS). During the cascade operation, only one transmitting device (e.g., AP) may perform downlink communication, and a plurality of transmitting devices (e.g., non-APs) may perform uplink communication.

(138) A CS request field **1230** indicates whether a wireless medium state or a NAV or the like is necessarily considered in a situation where a receiving device which has received a corresponding trigger frame transmits a corresponding uplink PPDU.

(139) An HE-SIG-A information field **1240** may include information for controlling content of a SIG-A field (i.e., HE-SIG-A field) of the uplink PPDU in response to the corresponding trigger frame.

(140) A CP and LTF type field **1250** may include information related to a CP length and LTF length of the uplink PPDU transmitted in response to the corresponding trigger frame. A trigger type field **1260** may indicate a purpose of using the corresponding trigger frame, for example, typical triggering, triggering for beamforming, a request for block ACK/NACK, or the like.

(141) It may be assumed that the trigger type field **1260** of the trigger frame in the present specification indicates a trigger frame of a basic type for typical triggering. For example, the trigger frame of the basic type may be referred to as a basic trigger frame.

(142) FIG. **13** illustrates an example of a subfield included in a per user information field. A user information field **1300** of FIG. **13** may be understood as any one of the per user information fields **1160#1** to **1160**#N mentioned above with reference to FIG. **11**. A subfield included in the user information field **1300** of FIG. **13** may be partially omitted, and an extra subfield may be added. In addition, a length of each subfield illustrated may be changed.

(143) A user identifier field **1310** of FIG. **13** indicates an identifier of a STA (i.e., receiving STA) corresponding to per user information. An example of the identifier may be the entirety or part of an association identifier (AID) value of the receiving STA.

(144) In addition, an RU allocation field **1320** may be included. That is, when the receiving STA identified through the user identifier field **1310** transmits a TB PPDU in response to the trigger frame, the TB PPDU is transmitted through an RU indicated by the RU allocation field **1320**. In this case, the RU indicated by the RU allocation field **1320** may be an RU shown in FIG. **5**, FIG. **6**, and FIG. **7**.

(145) The subfield of FIG. **13** may include a coding type field **1330**. The coding type field **1330** may indicate a coding type of the TB PPDU. For example, when BCC coding is applied to the TB PPDU, the coding type field **1330** may be set to ‘1’, and when LDPC coding is applied, the coding type field **1330** may be set to ‘0’.

(146) In addition, the subfield of FIG. **13** may include an MCS field **1340**. The MCS field **1340** may indicate an MCS scheme applied to the TB PPDU. For example, when BCC coding is applied to the TB PPDU, the coding type field **1330** may be set to ‘1’, and when LDPC coding is applied, the coding type field **1330** may be set to ‘0’.

(147) Hereinafter, a UL OFDMA-based random access (UORA) scheme will be described.

(148) FIG. **14** describes a technical feature of the UORA scheme.

(149) A transmitting STA (e.g., an AP) may allocate six RU resources through a trigger frame as shown in FIG. **14**. Specifically, the AP may allocate a 1st RU resource (AID 0, RU 1), a 2nd RU resource (AID 0, RU 2), a 3rd RU resource (AID 0, RU 3), a 4th RU resource (AID 2045, RU 4), a 5th RU resource (AID 2045, RU 5), and a 6th RU resource (AID 3, RU 6). Information related to the AID 0, AID 3, or AID 2045 may be included, for example, in the user identifier field **1310** of FIG. **13**. Information related to the RU 1 to RU 6 may be included, for example, in the RU allocation field **1320** of FIG. **13**. AID=0 may imply a UORA resource for an associated STA, and AID=2045 may imply a UORA resource for an un-associated STA. Accordingly, the 1st to 3rd RU

resources of FIG. **14** may be used as a UORA resource for the associated STA, the 4th and 5th RU resources of FIG. **14** may be used as a UORA resource for the un-associated STA, and the 6th RU resource of FIG. **14** may be used as a typical resource for UL MU.

(150) In the example of FIG. **14**, an OFDMA random access backoff (OBO) of a STA1 is decreased to 0, and the STA1 randomly selects the 2nd RU resource (AID 0, RU 2). In addition, since an OBO counter of a STA2/3 is greater than 0, an uplink resource is not allocated to the STA2/3. In addition, regarding a STA4 in FIG. **14**, since an AID (e.g., AID=3) of the STA4 is included in a trigger frame, a resource of the RU 6 is allocated without backoff.

(151) Specifically, since the STA1 of FIG. **14** is an associated STA, the total number of eligible RA RUs for the STA1 is 3 (RU 1, RU 2, and RU 3), and thus the STA1 decreases an OBO counter by 3 so that the OBO counter becomes 0. In addition, since the STA2 of FIG. **14** is an associated STA, the total number of eligible RA RUs for the STA2 is 3 (RU 1, RU 2, and RU 3), and thus the STA2 decreases the OBO counter by 3 but the OBO counter is greater than 0. In addition, since the STA3 of FIG. **14** is an un-associated STA, the total number of eligible RA RUs for the STA3 is 2 (RU 4, RU 5), and thus the STA3 decreases the OBO counter by 2 but the OBO counter is greater than 0.

(152) FIG. **15** illustrates an example of a channel used/supported/defined within a 2.4 GHz band.

(153) The 2.4 GHz band may be called in other terms such as a first band. In addition, the 2.4 GHz band may imply a frequency domain in which channels of which a center frequency is close to 2.4 GHz (e.g., channels of which a center frequency is located within 2.4 to 2.5 GHz) are used/supported/defined.

(154) A plurality of 20 MHz channels may be included in the 2.4 GHz band. 20 MHz within the 2.4 GHz may have a plurality of channel indices (e.g., an index 1 to an index 14). For example, a center frequency of a 20 MHz channel to which a channel index 1 is allocated may be 2.412 GHz, a center frequency of a 20 MHz channel to which a channel index 2 is allocated may be 2.417 GHz, and a center frequency of a 20 MHz channel to which a channel index N is allocated may be (2.407+0.005*N) GHz. The channel index may be called in various terms such as a channel number or the like. Specific numerical values of the channel index and center frequency may be changed.

(155) FIG. **15** exemplifies 4 channels within a 2.4 GHz band. Each of 1st to 4th frequency domains **1510** to **1540** shown herein may include one channel. For example, the 1st frequency domain **1510** may include a channel 1 (a 20 MHz channel having an index 1). In this case, a center frequency of the channel 1 may be set to 2412 MHz. The 2nd frequency domain **1520** may include a channel 6. In this case, a center frequency of the channel 6 may be set to 2437 MHz. The 3rd frequency domain **1530** may include a channel 11. In this case, a center frequency of the channel 11 may be set to 2462 MHz. The 4th frequency domain **1540** may include a channel 14. In this case, a center frequency of the channel 14 may be set to 2484 MHz.

(156) FIG. **16** illustrates an example of a channel used/supported/defined within a 5 GHz band.

(157) The 5 GHz band may be called in other terms such as a second band or the like. The 5 GHz band may imply a frequency domain in which channels of which a center frequency is greater than or equal to 5 GHz and less than 6 GHz (or less than 5.9 GHz) are used/supported/defined. Alternatively, the 5 GHz band may include a plurality of channels between 4.5 GHz and 5.5 GHz. A specific numerical value shown in FIG. **16** may be changed.

(158) A plurality of channels within the 5 GHz band include an unlicensed national information infrastructure (UNII)-1, a UNII-2, a UNII-3, and an ISM. The INII-1 may be called UNII Low. The UNII-2 may include a frequency domain called UNII Mid and UNII-2 Extended.

(159) The UNII-3 may be called UNII-Upper.

(160) A plurality of channels may be configured within the 5 GHz band, and a bandwidth of each channel may be variously set to, for example, 20 MHz, 40 MHz, 80 MHz, 160 MHz, or the like. For example, 5170 MHz to 5330 MHz frequency domains/ranges within the UNII-1 and UNII-2 may be divided into eight 20 MHz channels. The 5170 MHz to 5330 MHz frequency

domains/ranges may be divided into four channels through a 40 MHz frequency domain. The 5170 MHz to 5330 MHz frequency domains/ranges may be divided into two channels through an 80 MHz frequency domain. Alternatively, the 5170 MHz to 5330 MHz frequency domains/ranges may be divided into one channel through a 160 MHz frequency domain.
(161) FIG. **17** illustrates an example of a channel used/supported/defined within a 6 GHz band.
(162) The 6 GHz band may be called in other terms such as a third band or the like. The 6 GHz band may imply a frequency domain in which channels of which a center frequency is greater than or equal to 5.9 GHz are used/supported/defined. A specific numerical value shown in FIG. **17** may be changed.
(163) For example, the 20 MHz channel of FIG. **17** may be defined starting from 5.940 GHz. Specifically, among 20 MHz channels of FIG. **17**, the leftmost channel may have an index 1 (or a channel index, a channel number, etc.), and 5.945 GHz may be assigned as a center frequency. That is, a center frequency of a channel of an index N may be determined as (5.940+0.005*N) GHz.
(164) Accordingly, an index (or channel number) of the 2 MHz channel of FIG. **17** may be 1, 5, 9, 13, 17, 21, 25, 29, 33, 37, 41, 45, 49, 53, 57, 61, 65, 69, 73, 77, 81, 85, 89, 93, 97, 101, 105, 109, 113, 117, 121, 125, 129, 133, 137, 141, 145, 149, 153, 157, 161, 165, 169, 173, 177, 181, 185, 189, 193, 197, 201, 205, 209, 213, 217, 221, 225, 229, 233. In addition, according to the aforementioned (5.940+0.005*N) GHz rule, an index of the 40 MHz channel of FIG. **17** may be 3, 11, 19, 27, 35, 43, 51, 59, 67, 75, 83, 91, 99, 107, 115, 123, 131, 139, 147, 155, 163, 171, 179, 187, 195, 203, 211, 219, 227.
(165) Although 20, 40, 80, and 160 MHz channels are illustrated in the example of FIG. **17**, a 240 MHz channel or a 320 MHz channel may be additionally added.
(166) Hereinafter, a PPDU transmitted/received in a STA of the present specification will be described.
(167) FIG. **18** illustrates an example of a PPDU used in the present specification.
(168) The PPDU **1800** depicted in FIG. **18** may be referred to as various terms such as an EHT PPDU, a TX PPDU, an RX PPDU, a first type or N-th type PPDU, or the like. In addition, the EHT PPDU may be used in an EHT system and/or a new WLAN system enhanced from the EHT system.
(169) The subfields **1801** to **1810** depicted in FIG. **18** may be referred to as various terms. For example, a SIG A field **1805** may be referred to an EHT-SIG-A field, a SIG B field **1806** may be referred to an EHT-SIG-B, an STF field **1807** may be referred to an EHT-STF field, and an LTF field **1808** may be referred to an EHT-LTF.
(170) The subcarrier spacing of the L-LTF, L-STF, L-SIG, and RL-SIG fields **1801**, **1802**, **1803**, and **1804** of FIG. **18** can be set to 312.5 kHz, and the subcarrier spacing of the STF, LTF, and Data fields **1807**, **1808**, and **1809** of FIG. **18** can be set to 78.125 kHz. That is, the subcarrier index of the L-LTF, L-STF, L-SIG, and RL-SIG fields **1801**, **1802**, **1803**, and **1804** can be expressed in unit of 312.5 kHz, and the subcarrier index of the STF, LTF, and Data fields **1807**, **1808**, and **1809** can be expressed in unit of 78.125 kHz.
(171) The SIG A and/or SIG B fields of FIG. **18** may include additional fields (e.g., a SIG C field or one control symbol, etc.). The subcarrier spacing of all or part of the SIG A and SIG B fields may be set to 312.5 kHz, and the subcarrier spacing of all or part of newly-defined SIG field(s) may be set to 312.5 kHz. Meanwhile, the subcarrier spacing for a part of the newly-defined SIG field(s) may be set to a pre-defined value (e.g., 312.5 kHz or 78.125 kHz).
(172) In the PPDU of FIG. **18**, the L-LTF and the L-STF may be the same as conventional L-LTF and L-STF fields.
(173) The L-SIG field of FIG. **18** may include, for example, bit information of 24 bits. For example, the 24-bit information may include a rate field of 4 bits, a reserved bit of 1 bit, a length field of 12 bits, a parity bit of 1 bit, and a tail bit of 6 bits. For example, the length field of 12 bits may include information related to the number of octets of a corresponding Physical Service Data

Unit (PSDU). For example, the length field of 12 bits may be determined based on a type of the PPDU. For example, when the PPDU is a non-HT, HT, VHT PPDU or an EHT PPDU, a value of the length field may be determined as a multiple of 3. For example, when the PPDU is an HE PPDU, the value of the length field may be determined as “a multiple of 3”+1 or “a multiple of 3”+2. In other words, for the non-HT, HT, VHT PPDI or the EHT PPDU, the value of the length field may be determined as a multiple of 3, and for the HE PPDU, the value of the length field may be determined as “a multiple of 3”+1 or “a multiple of 3”+2.

(174) For example, the transmitting STA may apply BCC encoding based on a 1/2 coding rate to the 24-bit information of the L-SIG field. Thereafter, the transmitting STA may obtain a BCC coding bit of 48 bits. BPSK modulation may be applied to the 48-bit coding bit, thereby generating 48 BPSK symbols. The transmitting STA may map the 48 BPSK symbols to positions except for a pilot subcarrier {subcarrier index −21, −7, +7, +21} and a DC subcarrier {subcarrier index 0}. As a result, the 48 BPSK symbols may be mapped to subcarrier indices −26 to −22, −20 to −8, −6 to −1, +1 to +6, +8 to +20, and +22 to +26. The transmitting STA may additionally map a signal of {−1, −1, −1, 1} to a subcarrier index {−28, −27, +27, +28}. The aforementioned signal may be used for channel estimation on a frequency domain corresponding to {−28, −27, +27, +28}.

(175) The transmitting STA may generate an RL-SIG which is identical to the L-SIG. BPSK modulation may be applied to the RL-SIG. The receiving STA may figure out that the RX PPDU is the HE PPDU or the EHT PPDU, based on the presence of the RL-SIG.

(176) After RL-SIG of FIG. **18**, for example, an EHT-SIG-A or one control symbol may be inserted. A symbol located after the RL-SIG (i.e., the EHT-SIG-A or one control symbol in the present specification) may be referred as various names, such as a U-SIG (Universal SIG) field.

(177) A symbol contiguous to the RL-SIG (e.g., U-SIG) may include information of N bits, and may include information for identifying the type of the EHT PPDU. For example, the U-SIG may be configured based on two symbols (e.g., two contiguous OFDM symbols). Each symbol (e.g., OFDM symbol) for U-SIG may have a duration of 4 us. Each symbol of the U-SIG may be used to transmit 26-bit information. For example, each symbol of the U-SIG may be transmitted/received based on 52 data tones and 4 pilot tones.

(178) Through the U-SIG (or U-SIG field), for example, A-bit information (e.g., 52 un-coded bits) may be transmitted. A first symbol of the U-SIG may transmit first X-bit information (e.g., 26 un-coded bits) of the A-bit information, and a second symbol of the U-SIG may transmit the remaining Y-bit information (e.g. 26 un-coded bits) of the A-bit information. For example, the transmitting STA may obtain 26 un-coded bits included in each U-SIG symbol. The transmitting STA may perform convolutional encoding (i.e., BCC encoding) based on a rate of R=1/2 to generate 52-coded bits, and may perform interleaving on the 52-coded bits. The transmitting STA may perform BPSK modulation on the interleaved 52-coded bits to generate 52 BPSK symbols to be allocated to each U-SIG symbol. One U-SIG symbol may be transmitted based on 65 tones (subcarriers) from a subcarrier index −28 to a subcarrier index +28, except for a DC index 0. The 52 BPSK symbols generated by the transmitting STA may be transmitted based on the remaining tones (subcarriers) except for pilot tones, i.e., tones −21, −7, +7, +21.

(179) For example, the A-bit information (e.g., 52 un-coded bits) generated by the U-SIG may include a CRC field (e.g., a field having a length of 4 bits) and a tail field (e.g., a field having a length of 6 bits). The CRC field and the tail field may be transmitted through the second symbol of the U-SIG. The CRC field may be generated based on 26 bits allocated to the first symbol of the U-SIG and the remaining 16 bits except for the CRC/tail fields in the second symbol, and may be generated based on the conventional CRC calculation algorithm. In addition, the tail field may be used to terminate trellis of a convolutional decoder, and may be set to, for example, “000000”.

(180) The A-bit information (e.g., 52 un-coded bits) transmitted by the U-SIG (or U-SIG field) may be divided into version-independent bits and version-dependent bits. For example, the version-independent bits may have a fixed or variable size. For example, the version-independent bits may

be allocated only to the first symbol of the U-SIG, or the version-independent bits may be allocated to both of the first and second symbols of the U-SIG. For example, the version-independent bits and the version-dependent bits may be called in various terms such as a first control bit, a second control bit, or the like.

(181) For example, the version-independent bits of the U-SIG may include a PHY version identifier of 3 bits. For example, the PHY version identifier of 3 bits may include information related to a PHY version of a TX/RX PPDU. For example, a first value of the PHY version identifier of 3 bits may indicate that the TX/RX PPDU is an EHT PPDU. In other words, when the transmitting STA transmits the EHT PPDU, the PHY version identifier of 3 bits may be set to a first value. In other words, the receiving STA may determine that the RX PPDU is the EHT PPDU, based on the PHY version identifier having the first value.

(182) For example, the version-independent bits of the U-SIG may include a UL/DL flag field of 1 bit. A first value of the UL/DL flag field of 1 bit relates to UL communication, and a second value of the UL/DL flag field relates to DL communication.

(183) For example, the version-independent bits of the U-SIG may include information related to a TXOP length and information related to a BSS color ID.

(184) For example, when the EHT PPDU is classified into various types (e.g., EHT PPDU supporting SU, EHT PPDU supporting MU, EHT PPDU related to Trigger Frame, EHT PPDU related to Extended Range transmission, etc.), information related to the type of the EHT PPDU may be included in version-independent bits or version-dependent bits of the U-SIG.

(185) For example, the U-SIG field includes 1) a bandwidth field including information related to a bandwidth, 2) a field including information related an MCS scheme applied to the SIG-B, 3) a dual subcarrier modulation in the SIG-B (i.e., an indication field including information related to whether the dual subcarrier modulation) is applied, 4) a field including information related to the number of symbols used for the SIG-B, 5) a field including information on whether the SIG-B is generated over the entire band, 6) a field including information related to a type of the LTF/STF, and/or 7) information related to a field indicating a length of the LTF and the CP.

(186) The SIG-B of FIG. **18** may include the technical features of HE-SIG-B shown in the example of FIGS. **8** to **9** as it is.

(187) An STF of FIG. **18** may be used to improve automatic gain control estimation in a multiple input multiple output (MIMO) environment or an OFDMA environment. An LTF of FIG. **18** may be used to estimate a channel in the MIMO environment or the OFDMA environment.

(188) The EHT-STF of FIG. **18** may be set in various types. For example, a first type of STF (e.g., 1×STF) may be generated based on a first type STF sequence in which a non-zero coefficient is arranged with an interval of 16 subcarriers. An STF signal generated based on the first type STF sequence may have a period of 0.8p, and a periodicity signal of 0.8 μs may be repeated 5 times to become a first type STF having a length of 4 μs. For example, a second type of STF (e.g., 2×STF) may be generated based on a second type STF sequence in which a non-zero coefficient is arranged with an interval of 8 subcarriers. An STF signal generated based on the second type STF sequence may have a period of 1.6p, and a periodicity signal of 1.6 μs may be repeated 5 times to become a second type STF having a length of 8 μs. For example, a third type of STF (e.g., 4×STF) may be generated based on a third type STF sequence in which a non-zero coefficient is arranged with an interval of 4 subcarriers. An STF signal generated based on the third type STF sequence may have a period of 3.2 μs, and a periodicity signal of 3.2 μs may be repeated 5 times to become a second type STF having a length of 16 μs. Only some of the first to third type EHT-STF sequences may be used. In addition, the EHT-LTF field may also have first, second, and third types (i.e., 1×, 2×, 4×LTF). For example, the first/second/third type LTF field may be generated based on an LTF sequence in which a non-zero coefficient is arranged with an interval of 4/2/1 subcarriers. The first/second/third type LTF may have a time length of 3.2/6.4/12.8 μs. In addition, Guard Intervals (GIs) with various lengths (e.g., 0.8/1/6/3.2 μs) may be applied to the first/second/third type LTF.

(189) Information related to the type of STF and/or LTF (including information related to GI applied to the LTF) may be included in the SIG A field and/or the SIG B field of FIG. **18**.
(190) The PPDU of FIG. **18** may support various bandwidths. For example, the PPDU of FIG. **18** may have a bandwidth of 20/40/80/160/240/320 MHz. For example, at least one field (e.g., STF, LTF, data) of FIG. **18** may be configured based on RUs illustrated in FIGS. **5** to **7**, and the like. For example, when there is one receiving STA of the PPDU of FIG. **18**, all fields of the PPDU of FIG. **18** may occupy the entire bandwidth. For example, when there are multiple receiving STAs of the PPDU of FIG. **18** (i.e., when MU PPDU is used), some fields (e.g., STF, LTF, data) of FIG. **18** may be configured based on the RUs shown in FIGS. **5** to **7**. For example, the STF, LTF, and data fields for the first receiving STA of the PPDU may be transmitted/received through a first RU, and the STF, LTF, and data fields for the second receiving STA of the PPDU may be transmitted/received through a second RU. In this case, the locations/positions of the first and second RUs may be determined based on FIGS. **5** to **7**, and the like.
(191) The PPDU of FIG. **18** may be determined (or identified) as an EHT PPDU based on the following method.
(192) A receiving STA may determine a type of an RX PPDU as the EHT PPDU, based on the following aspect. For example, the RX PPDU may be determined as the EHT PPDU: 1) when a first symbol after an L-LTF signal of the RX PPDU is a BPSK symbol; 2) when RL-SIG in which the L-SIG of the RX PPDU is repeated is detected; and 3) when a result of applying "modulo 3" to a value of a length field of the L-SIG of the RX PPDU is detected as "0". When the RX PPDU is determined as the EHT PPDU, the receiving STA may detect a type of the EHT PPDU (e.g., an SU/MU/Trigger-based/Extended Range type), based on bit information included in a symbol after the RL-SIG of FIG. **18**. In other words, the receiving STA may determine the RX PPDU as the EHT PPDU, based on: 1) a first symbol after an L-LTF signal, which is a BPSK symbol; 2) RL-SIG contiguous to the L-SIG field and identical to L-SIG; 3) L-SIG including a length field in which a result of applying "modulo 3" is set to "0"; and 4) a 3-bit PHY version identifier of the aforementioned U-SIG (e.g., a PHY version identifier having a first value).
(193) For example, the receiving STA may determine the type of the RX PPDU as the EHT PPDU, based on the following aspect. For example, the RX PPDU may be determined as the HE PPDU: 1) when a first symbol after an L-LTF signal is a BPSK symbol; 2) when RL-SIG in which the L-SIG is repeated is detected; and 3) when a result of applying "modulo 3" to a value of a length field of the L-SIG is detected as "1" or "2".
(194) For example, the receiving STA may determine the type of the RX PPDU as a non-HT, HT, and VHT PPDU, based on the following aspect. For example, the RX PPDU may be determined as the non-HT, HT, and VHT PPDU: 1) when a first symbol after an L-LTF signal is a BPSK symbol; and 2) when RL-SIG in which L-SIG is repeated is not detected. In addition, even if the receiving STA detects that the RL-SIG is repeated, when a result of applying "modulo 3" to the length value of the L-SIG is detected as "0", the RX PPDU may be determined as the non-HT, HT, and VHT PPDU.
(195) In the following example, a signal represented as a (TX/RX/UL/DL) signal, a (TX/RX/UL/DL) frame, a (TX/RX/UL/DL) packet, a (TX/RX/UL/DL) data unit, (TX/RX/UL/DL) data, or the like may be a signal transmitted/received based on the PPDU of FIG. **18**. The PPDU of FIG. **18** may be used to transmit/receive frames of various types. For example, the PPDU of FIG. **18** may be used for a control frame. An example of the control frame may include a request to send (RTS), a clear to send (CTS), a power save-poll (PS-poll), BlockACKReq, BlockAck, a null data packet (NDP) announcement, and a trigger frame. For example, the PPDU of FIG. **18** may be used for a management frame. An example of the management frame may include a beacon frame, a (re-)association request frame, a (re-)association response frame, a probe request frame, and a probe response frame. For example, the PPDU of FIG. **18** may be used for a data frame. For example, the PPDU of FIG. **18** may be used to simultaneously transmit at least two or more of the

control frame, the management frame, and the data frame.

(196) FIG. **19** illustrates an example of a modified transmission device and/or receiving device of the present specification.

(197) Each device/STA of the sub-figure (a)/(b) of FIG. **1** may be modified as shown in FIG. **19**. A transceiver **630** of FIG. **19** may be identical to the transceivers **113** and **123** of FIG. **1**. The transceiver **630** of FIG. **19** may include a receiver and a transmitter.

(198) A processor **610** of FIG. **19** may be identical to the processors **111** and **121** of FIG. **1**. Alternatively, the processor **610** of FIG. **19** may be identical to the processing chips **114** and **124** of FIG. **1**.

(199) A memory **620** of FIG. **19** may be identical to the memories **112** and **122** of FIG. **1**. Alternatively, the memory **620** of FIG. **19** may be a separate external memory different from the memories **112** and **122** of FIG. **1**.

(200) Referring to FIG. **19**, a power management module **611** manages power for the processor **610** and/or the transceiver **630**. A battery **612** supplies power to the power management module **611**. A display **613** outputs a result processed by the processor **610**. A keypad **614** receives inputs to be used by the processor **610**. The keypad **614** may be displayed on the display **613**. A SIM card **615** may be an integrated circuit which is used to securely store an international mobile subscriber identity (IMSI) and its related key, which are used to identify and authenticate subscribers on mobile telephony devices such as mobile phones and computers.

(201) Referring to FIG. **19**, a speaker **640** may output a result related to a sound processed by the processor **610**. A microphone **641** may receive an input related to a sound to be used by the processor **610**.

(202) Hereinafter, a method for performing low-latency communication may be described. Low-latency communication may refer to technology for supporting time delay-sensitive traffic (that is, low-latency traffic). The time delay may mean latency defined in the IEEE 802.11ax standard.

(203) For example, the time delay (that is, latency) may mean a time from a time point at which a frame is received by the queue of the MAC layer to a time point at which the frame is deleted from the queue of the MAC layer. Specifically, the frame may come into the queue of the MAC layer of the transmitting STA (for example, AP). Thereafter, the frame may be transmitted through the PHY layer of the transmitting STA. The frame may be successfully received by the receiving STA. The transmitting STA may receive an ACK/Block ACK frame, or the like from the receiving STA. The transmitting STA may delete the frame from the queue of the MAC layer. Here, the time delay may mean a time from when a frame is received by the queue of the MAC layer until the frame is deleted from the queue of the MAC layer. Hereinafter, for the convenience of description, the transmitting STA may be referred to as an access point (AP). Also, the receiving STA may be referred to as a STA.

(204) Various technologies may be required to support time delay-sensitive traffic. For example, a method for transmitting a low latency frame, a method for low latency retransmission, a method for a low latency channel contention, or a method for low latency signaling may be required.

(205) As an example, the method for the low latency frame transmission (or technology for low latency communication) may mean technology for reducing the time delay when transmitting a data frame. As another example, the method for low latency retransmission may refer to a technique for reducing a time delay when the data frame is retransmitted, if transmission of the data frame fails. As another example, the method for low latency channel contention may refer to a technique for reducing the time by improving the channel contention method. As another example, the method for low latency signaling may refer to a signaling technology for exchanging information related to low latency communication between the STA and the AP to perform low latency communication.

(206) Hereinafter, the present specification may propose various technologies for supporting traffic sensitive to the above-described time delay. In addition, the traffic may include various types of

traffic. For example, traffic may be divided into at least two types of traffic. As an example, the first traffic may be traffic sensitive to time delay. The second traffic may be traffic that is not sensitive to time delay. Classification of traffic according to time delay may be only one example, and classification criteria may be set in various ways. For example, the classification criteria may include at least one-time delay, whether it is for machine-type communication, or importance.
(207) Hereinafter, traffic described in this specification may refer to a type of traffic different from conventional traffic. For example, the traffic described below may mean traffic sensitive to time delay.
(208) FIG. **20** is a flowchart related to an operation for performing low latency communication.
(209) Referring to FIG. **20**, the AP and the STA may perform steps S**2010** to S**2060**. Some of the steps described above may not be essential. Accordingly, some steps may be omitted. In addition, since the order of the above-described steps is an example, the order of performing each step may vary. In addition, only one of the above-described steps may have an independent technical meaning.
(210) In step S**2010**, the AP and the STA may perform an association process. Specifically, the AP and the STA may exchange information about the low-latency function. Information on the low latency function may include information on whether or not to support the low latency communication function.
(211) In step S**2020**, the AP and the STA may transmit/receive a low latency communication request frame and transmit/receive a low latency communication response frame. For example, uplink traffic may occur in the STA. The STA may request the AP to perform low latency communication based on the uplink traffic. That is, the STA may transmit a low latency communication request frame to the AP. AP may transmit a low latency communication response frame to the STA based on the low latency communication request frame.
(212) As another example, downlink traffic may be generated from the AP. The AP may transmit information to the STA that low latency communication will be performed, based on the downlink traffic. That is, the AP may transmit a low latency communication request frame to the STA. The STA may transmit a low latency communication response frame to the AP, based on the low latency communication request frame. In the case of downlink traffic, even when the STA does not transmit a low latency communication response frame to the AP, the AP and the STA may perform low latency communication.
(213) In step S**2030**, the AP and the STA may perform low-latency communication. The AP and the STA may transmit/receive traffic sensitive to time delay (that is, low-latency traffic).
(214) In step S**2040**, the AP and the STA may exchange time latency information (or information on time latency). The AP and the STA may exchange time latency information of current traffic while low latency communication is performed. For example, the AP and the STA may exchange the current traffic time latency status. The AP and the STA may periodically exchange time latency information, alternatively, the AP and the STA may exchange time latency information whenever it is necessary.
(215) In step S**2050**, the AP and the STA may change a specific operation for low-latency communication. The AP and the STA may change or maintain a specific operation for low-latency communication based on the current time latency.
(216) In step S**2060**, the AP and the STA may terminate the low latency communication. Based on all the traffic transmitted through the low latency communication is transmitted, the AP and the STA may request and respond to terminate the low latency communication.
(217) FIG. **21** is a diagram illustrating four queues of EDCA.
(218) Currently, IEEE 802.11 operates 4 queues for each STA. Referring to FIG. **21**, the STA may operate 4 queues. A queue may be referred to as an access category. An access category for voice may be referred to as AC_VO, an access category for video may be referred to as AC_VO, an access category for best effort may be referred to as AC_BE, and an access category for

background may be referred to as AC_BK. AC_VO, AC_VI, AC_BE, AC_BK may have priority in channel access in the order. The priority may be determined through an enhanced distributed channel access (EDCA) parameter. An access point (AP) may determine an EDCA parameter and inform the STA (for example, a non-AP STA). The STA must operate according to the determined EDCA parameter. When a MAC service data unit (MSDU) arrives at the MAC layer, the priority of the MSDU (that is, an access category) may be determined according to a User Priority function (UP). The MSDU for which the access category is determined may become a MAC protocol data unit (MPDU) by adding a MAC header. The MPDU may enter a queue for each corresponding access category and can access a channel according to the EDCA parameter determined for each queue.
(219) AC_VO and AC_VI may have priority over other access categories (for example, AC_BE and AC_BK) in order to support latency-important traffic such as voice and video. However, according to the User Priority function (UP), even for traffic that is not sensitive to latency, the access category may be determined as AC_VO or AC_VI according to the determination of the STA. A STA may use the same EDCA parameters as other STAs, and may have to perform contention in channels between AC_VOs and AC_VIs.
(220) According to an embodiment, the STA may receive traffic data (for example, MAC service data unit (MSDU)) having a pre-configured user priority from a higher layer.
(221) For example, in order to determine the transmission order of MAC frames to be transmitted by the STA, a differential value may be configured for each traffic data in the user priority. The user priority may be mapped to each access category (AC) in which traffic data is buffered in a manner as shown in Table 5 below.
(222) TABLE-US-00005 TABLE 5 Priority User Priority Access Category (AC) Lower 1 AC_BK 2 AC_BK 0 AC_BE 3 AC_BE 4 AC_VI 5 AC_VI 6 AC_VO Higher 7 AC_VO
(223) In this specification, the user priority may be understood as a traffic identifier (hereinafter, 'TID') indicating characteristics of traffic data.
(224) Referring to Table 5, traffic data having a user priority (that is, TID) of '1' or '2' may be buffered in the AC_BK type transmission queue **2150**. Traffic data having a user priority (that is, TID) of '0' or '3' may be buffered in the AC_BE type transmission queue **2140**.
(225) Traffic data having a user priority (that is, TID) of '4' or '5' may be buffered in the AC_VI type transmission queue **2130**. Traffic data having a user priority (that is, TID) of '6' or '7' may be buffered in the AC_VO type transmission queue **2120**.
(226) Instead of DIFS (DCF interframe space), CWmin, CWmax, which are parameters for backoff operation/procedure based on the existing distributed coordination function (DCF), EDCA parameter, configured for the backoff operation/procedure of the STA performing EDCA, AIFS (arbitration interframe space) [AC], CWmin [AC], CWmax [AC] and TXOP limit [AC] may be used.
(227) A difference in transmission priority between ACs may be implemented based on the differential EDCA parameter set. The default values of the EDCA parameter sets (that is, AIFS[AC], CWmin[AC], CWmax[AC], TXOP limit[AC]) corresponding to each AC are exemplarily shown in Table 6 below. Specific values in Table 6 could be configured differently from the following.
(228) TABLE-US-00006 TABLE 6 AC CWmin[AC] CWmax[AC] AIFS[AC] TXOP limit[AC] AC_BK 31 1023 7 0 AC_BE 31 1023 3 0 AG_VI 15 31 2 3.008 ms AG_VO 7 15 2 1.504 ms
(229) The EDCA parameter set for each AC may be configured as a default value or may be included in a beacon frame and transmitted from an access point (AP) to each STA. The smaller the AIFS[AC] and CWmin[AC] values, the higher the priority. As a result, channel access delay is shortened, allowing more bandwidth to be used in a given traffic environment.
(230) The EDCA parameter set may include information on channel access parameters (for example, AIFS[AC], CWmin[AC], CWmax[AC]) for each AC.

(231) The backoff operation/procedure for EDCA may be performed based on an EDCA parameter set individually configured to four ACs included in each STA. Appropriate configuring of EDCA parameter values, which define different channel access parameters for each AC, can optimize network performance and increase the transmission effect due to traffic priority.
(232) Therefore, the AP of the WLAN system must perform overall management and adjustment functions for EDCA parameters to ensure fair media access to all STAs participating in the network.
(233) Referring to FIG. **21**, one STA (or AP) **2100** may include a virtual mapper **2110**, a plurality of transmission queues **2120** to **2150**, and a virtual collision handler **2160**. The virtual mapper **2110** of FIG. **21** may serve to map the MSDU received from the logical link control (LLC) layer to the transmission queue corresponding to each AC according to Table 1 above.
(234) The plurality of transmission queues **2120** to **2150** of FIG. **21** may serve as individual EDCA contention entities for wireless medium access within one STA (or AP).
(235) In the following specification, a technical feature for designating whether the AP uses each EDCA Queue of specific STAs may be proposed. A transmittable Access Category according to the aforementioned Access type may be described in Table 7.
(236) TABLE-US-00007 TABLE 7 Access type Access Category 1 VO only 2 VI only 3 BE only 4 BK only 5 VO, VI 6 BE, BK 7 VO, VI, BE, BK
(237) Referring to Table 7, a transmittable Access Category may be configured differently based on an Access type.
(238) For example, when the access type is configured as a first value (for example, 1), only AC_VO (or AC_VO type) traffic could be transmitted.
(239) For example, when the access type is configured as a second value (for example, 2), only AC_VI (or AC_VI type) traffic could be transmitted.
(240) For example, when the access type is configured as a third value (for example, 3), only AC_BE (or AC_BE type) traffic could be transmitted.
(241) For example, when the access type is configured as a fourth value (for example, 4), only AC_BK (or AC_BK type) traffic could be transmitted.
(242) For example, when the access type is configured as a fifth value (for example, 5), only traffic of AC_VO (or AC_VO type) and/or AC_VI (AC_VI type) could be transmitted.
(243) For example, when the access type is configured as a sixth value (for example, 6), only traffic of A AC_BE (or AC_BE type) and/or AC_BK (or AC_BK type) could be transmitted.
(244) For example, when the access type is configured as a seventh value (for example, 7), only traffic of AC_VO, AC_VI, AC_BE, and/or AC_BK could be transmitted.
(245) According to an embodiment, the AP may transmit information about the access type to the STA. Based on the information on the access type indicated by the AP, the access category that the STA can transmit may be limited. In other words, based on the information on the access type transmitted by the AP, the transmittable access category of the STA may be limited. For example, the AP may designate the access type as 1 to the first STA. The first STA may transmit only AC_VO (or AC_VO type traffic). As another example, the AP may transmit via broadcast a Beacon frame, and the like, after designating the access type as 5. In this case, only AC_VO and AC_VI may be transmitted in the corresponding BSS.
(246) According to an embodiment, the access type shown in Table 7 is an example and may be configured in various ways.
(247) Hereinafter, a method for the AP to indicate an access type configured as shown in Table 7 may be described. The AP may indicate the access type through various methods. In other words, the AP may transmit information about the access type through various methods.
(248) For example, the AP may indicate the access type through two methods.
(249) 1. A Method for Unicast Transmission
(250) 1-A. The unicast transmission method is a method in which the AP indicates an access type

to a specific STA using a unicast frame. In other words, the AP may transmit information about the access type to a specific STA through a unicast frame. 1-B. The AP may indicate the access type by using a Probe Response frame, an Association Response frame, or a specific Management frame. In other words, the unicast frame may include a Probe Response frame, an Association Response frame, or a specific Management frame. 1-C. The above-described frame (for example, Probe Response frame, Association Response frame, or specific Management frame, etc.) may include not only the Access Type but also the EDCA parameter of each Access category. 1-D. The STA that has received the above-described frame may transmit only a data frame for a determined access category. 1-E. In addition to the data frame, the AP may separately indicate whether a control frame or a management frame can be transmitted.
2. A Method for Broadcast Transmission 2-A. The broadcast transmission method is a method in which the AP transmits to all unspecified STAs using a broadcast frame. In other words, the AP may transmit information about the access type to all unspecified STAs through a broadcast frame. 2-B. According to an embodiment, the AP may periodically transmit an access type (or information on an access type) through a beacon frame. According to an embodiment, the AP may aperiodically transmit an access type (or information on an access type) through a broadcast frame in a specific situation (or a specified condition). 2-C. The above-described frame (for example, Broadcast frame) may include not only the Access Type but also the EDCA parameter of each Access category. 2-D. The STA that has received the above-described frame may transmit only a data frame for the determined access category. 2-E. In addition to the data frame, the AP may separately indicate whether a control frame or a management frame can be transmitted.
(251) FIG. **22** is a flowchart illustrating an operation of an STA.
(252) Referring to FIG. **22**, in step S**2210**, the STA may establish a connection with the AP. The STA and AP may support a low-latency communication function. For example, the STA may perform the operation shown in FIG. **3**. As an example, the STA may establish a connection with the AP based on at least one of steps S**310** to S**340** of FIG. **3**.
(253) According to an embodiment, the STA may receive low-latency communication capability information from the AP before establishing a connection with the AP. For example, the STA may receive information on whether to support a low-latency communication function (that is, low-latency communication capability information) through a Beacon frame or a Probe response frame. As an example, information on whether to support a low-latency communication function may be included in the EHT Capability information element field. According to an embodiment, the STA may transmit low-latency communication capability information to the AP. Step S**2210** may be related to step S**2010** of FIG. **20**.
(254) In step S**2220**, the STA may receive information related to an access type.
(255) According to an embodiment, the STA and the AP may perform low-latency communication. When performing low-latency communication, the STA may be limited in the access category of data that can be transmitted. The STA may identify an access category of data that can be transmitted based on the information about the access type.
(256) According to an embodiment, information on an access type may be configured as shown in Table 7 above. Based on the value of the access type, a transmittable access category may be determined.
(257) According to an embodiment, the STA may receive information about an access type through unicast. For example, the STA may receive information about an access type based on a Probe Response frame, an Association Response frame, a specific Management frame, or the like.
(258) According to an embodiment, the STA may receive information about an access type through broadcast. For example, the STA may receive information on an access type based on a Beacon frame, or the like.
(259) In step S**2230**, the STA may perform the indicated EDCA transmission. According to an embodiment, the STA may receive information about the EDCA parameter of each access category

from the AP. The STA may perform EDCA transmission based on the information on the EDCA parameter.
(260) According to an embodiment, the STA may select a backoff value for the backoff operation/procedure. In addition, the STA may attempt transmission after waiting for the selected backoff value for a time (that is, a backoff window) expressed in units of slot time. Also, the STA may count down the backoff window in units of slot times.
(261) According to an embodiment, the STA may transmit data related to a transmittable access category to the AP. For example, the STA may confirm that the transmittable access category is AC_VO based on information about the access type. The STA may transmit data (or traffic) regarding AC_VO to the AP.
(262) FIG. **23** is a flowchart for explaining the operation of the AP.
(263) Referring to FIG. **23**, in step S**2310**, the AP may establish a connection with the STA. The STA and AP may support a low-latency communication function. For example, the AP may perform the operation shown in FIG. **3**. As an example, the AP may establish a connection with the STA based on at least one of steps S**310** to S**340** of FIG. **3**.
(264) According to one embodiment, before establishing a connection with the STA, the AP may transmit low-latency communication capability information. For example, through a Beacon frame, or Probe response frame, or the like, the AP may transmit information about whether the low-latency communication function support (that is, low-latency communication capability information). As an example, information on whether to support a low-latency communication function may be included in the EHT Capability information element field. According to an embodiment, the AP may receive low-latency communication capability information from the STA. Step S**2310** may be related to step S**2010** of FIG. **20**.
(265) In step S**2320**, the AP may determine the access type. AP and STA may perform low-latency communication. AP may limit the access category that can be transmitted from the STA for the low-latency communication.
(266) According to an embodiment, information on an access type may be configured as shown in Table 7 above. Based on the value of the access type, an access category that can be transmitted by the STA may be determined.
(267) In step S**2330**, the AP may transmit information about the access type.
(268) According to an embodiment, the AP may transmit information about an access type via unicast. For example, the AP may transmit information about an access type based on a Probe Response frame, an Association Response frame, a specific Management frame, or the like.
(269) According to an embodiment, the AP may transmit information about an access type via a broadcast. For example, the AP may transmit information about an access type based on a Beacon frame or the like.
(270) According to an embodiment, the AP may transmit information about the EDCA parameter of each access category to the STA together. The STA may perform EDCA transmission based on the information on the EDCA parameter. Thereafter, the AP may receive data about an access category that the STA may transmit from the STA.
(271) FIG. **24** is a flowchart illustrating an operation of a receiving STA.
(272) Referring to FIG. **24**, in step S**2410**, a receiving STA may receive information about an access type. For example, the receiving STA may receive information about the access type from the transmitting STA.
(273) According to an embodiment, the receiving STA may perform low-latency communication with the transmitting STA. That is, the receiving STA and the transmitting STA may operate in a mode for low-latency communication. In the mode for the low-latency communication, the receiving STA may receive information about a transmittable access category from the transmitting STA, and transmit only data about the transmittable access category to the transmitting STA. For example, the data may include data for which a time delay value less than or equal to a threshold

value is required.
(274) According to an embodiment, the receiving STA may receive the information on the access type from the transmitting STA in unicast through a Probe Response frame or an Association Response frame.
(275) According to an embodiment, the receiving STA may receive information about the access type from the transmitting STA in a broadcast through a beacon frame.
(276) In step S**2420**, the receiving STA may check a transmittable access category.
(277) According to an embodiment, the information related to the access type may include information about the transmittable access category. A transmittable access category may be configured as at least one of AC_VO, AC_VI, AC_BE, and AC_BK.
(278) For example, the information on the access type may be configured as shown in Table 7 above.
(279) For example, based on the access type information being a first value (for example, 1), a transmittable access category may be set to AC_VO.
(280) As an example, based on the access type information being a second value (for example, 2), a transmittable access category may be set to AC_VI.
(281) For example, based on the access type information being a third value (for example, 3), a transmittable access category may be set to AC_BE.
(282) As an example, based on the access type information being a fourth value (for example, 4), a transmittable access category may be set to AC_BK.
(283) For example, based on the access type information being a fifth value (for example, 5), transmittable access categories may be set to AC_VO and AC_VI.
(284) For example, based on the access type information being a sixth value (for example, 6), transmittable access categories may be set to AC_BE and AC_BK.
(285) For example, based on the access type information being a seventh value (for example, 7), transmittable access categories may be set to AC_VO, AC_VI, AC_BE, and AC_BK.
(286) In step S**2430**, the receiving STA may transmit data related to the transmittable access category. For example, the receiving STA may transmit data regarding a transmittable access category to the transmitting STA.
(287) According to an embodiment, the receiving STA may receive information about the EDCA parameter of each access category from the transmitting STA. The receiving STA may perform EDCA transmission based on the information on the EDCA parameter.
(288) According to an embodiment, the receiving STA may transmit data related to a transmittable access category to the transmitting STA. For example, the receiving STA may confirm that the transmittable access category is AC_BK based on information about the access type. The receiving STA may transmit data (or traffic) regarding AC_BK to the transmitting STA.
(289) FIG. **25** is a flowchart illustrating an operation of a transmitting STA.
(290) Referring to FIG. **25**, in step S**2510**, a transmitting STA may determine information related to an access type.
(291) According to an embodiment, the receiving STA may perform low-latency communication with the transmitting STA. That is, the receiving STA and the transmitting STA may operate in a mode for low-latency communication. In the low-latency communication mode, the transmitting STA transmits information about the transmittable access category to the receiving STA, and may receive only data about the transmittable access category from the transmitting STA. For example, the data may include data for which a time delay value less than or equal to a threshold value is required.
(292) For example, the transmitting STA may confirm that the receiving STA is going to transmit data sensitive to time delay. The transmitting STA may check the access category of the data sensitive to the time delay. The transmitting STA may determine information about the access type so that the receiving STA can smoothly transmit the data sensitive to the time delay. As an example,

the transmitting STA may configure only the receiving STA to transmit data related to the access category. In addition, the transmitting STA may set a STA other than the receiving STA to not transmit data related to the access category.

(293) As an example, the transmitting STA may confirm that the receiving STA is going to transmit AC_VO data. The transmitting STA may determine information about the access type so that the receiving STA can smoothly transmit AC_VO data.

(294) According to an embodiment, the information related to the access type may include information about the transmittable access category. A transmittable access category may be configured as at least one of AC_VO, AC_VI, AC_BE, and AC_BK.

(295) For example, information about the access type may be set as shown in Table 7.

(296) For example, based on the access type information being a first value (for example, 1), a transmittable access category may be set to AC_VO.

(297) As an example, based on the access type information being a second value (for example, 2), a transmittable access category may be set to AC_VI.

(298) For example, based on the access type information being a third value (for example, 3), a transmittable access category may be set to AC_BE.

(299) As an example, based on the access type information being a fourth value (for example, 4), a transmittable access category may be set to AC_BK.

(300) For example, based on the access type information being a fifth value (for example, 5), transmittable access categories may be set to AC_VO and AC_VI.

(301) For example, based on the access type information being a sixth value (for example, 6), transmittable access categories may be set to AC_BE and AC_BK.

(302) For example, based on the access type information being a seventh value (for example, 7), transmittable access categories may be set to AC_VO, AC_VI, AC_BE, and AC_BK.

(303) In step S**2520**, the transmitting STA may transmit information related to the access type. For example, the transmitting STA may transmit information about the access type to the receiving STA.

(304) According to an embodiment, the transmitting STA may transmit the information on the access type to the receiving STA in a unicast manner through a Probe Response frame or an Association Response frame.

(305) According to an embodiment, the transmitting STA may transmit information about the access type to the receiving STA in a broadcast manner through a beacon frame.

(306) In step S**2530**, the transmitting STA may receive data. For example, the transmitting STA may receive data from the receiving STA based on the information on the access type. The data may include data about a transmittable access category.

(307) For example, the transmitting STA may receive data from the receiving STA regarding a transmittable access category. As an example, the transmitting STA may transmit that the transmittable access category is AC_BK to the receiving STA based on information about an access type. The transmitting STA may receive data (or traffic) regarding AC_BK from the receiving STA.

(308) The technical features of the present disclosure described above may be applied to various devices and methods. For example, the above-described technical features of the present disclosure may be performed/supported through the apparatus of FIGS. **1** and/or **19**. For example, the above-described technical features of the present disclosure may be applied only to a part of FIGS. **1** and/or **19**. For example, the technical features of the present disclosure described above may be implemented based on the processing chips **114** and **124** of FIG. **1**, may be implemented based on the processors **111** and **121** and the memories **112** and **122** of FIG. **1**, or may be implemented based on the processor **610** and the memory **620** of FIG. **19**. For example, the device of the present specification may comprise a processor; and a memory coupled to the processor, wherein the processor is configured to: receive, from a transmitting STA, information related to an access type; identify a transmittable access category, based on the information about the access type; and

transmit, to the transmitting STA, data related to the transmittable access category.
(309) The technical features of the present disclosure may be implemented based on a computer readable medium (CRM). For example, a CRM proposed by the present disclosure may store instructions which perform operations including the steps of acquiring, from a transmitting STA, information related to an access type; identifying a transmittable access category, based on the information about the access type; and transmitting, to the transmitting STA, data related to the transmittable access category. The instructions stored in the CRM of the present disclosure may be executed by at least one processor. At least one processor related to CRM in the present disclosure may be the processors **111** and **121** or the processing chips **114** and **124** of FIG. **1**, or the processor **610** of FIG. **19**. Meanwhile, the CRM of the present disclosure may be the memories **112** and **122** of FIG. **1**, the memory **620** of FIG. **19**, or a separate external memory/storage medium/disk.
(310) The foregoing technical features of the present specification are applicable to various applications or business models. For example, the foregoing technical features may be applied for wireless communication of a device supporting artificial intelligence (AI).
(311) Artificial intelligence refers to a field of study on artificial intelligence or methodologies for creating artificial intelligence, and machine learning refers to a field of study on methodologies for defining and solving various issues in the area of artificial intelligence. Machine learning is also defined as an algorithm for improving the performance of an operation through steady experiences of the operation.
(312) An artificial neural network (ANN) is a model used in machine learning and may refer to an overall problem-solving model that includes artificial neurons (nodes) forming a network by combining synapses. The artificial neural network may be defined by a pattern of connection between neurons of different layers, a learning process of updating a model parameter, and an activation function generating an output value.
(313) The artificial neural network may include an input layer, an output layer, and optionally one or more hidden layers. Each layer includes one or more neurons, and the artificial neural network may include synapses that connect neurons. In the artificial neural network, each neuron may output a function value of an activation function of input signals input through a synapse, weights, and deviations.
(314) A model parameter refers to a parameter determined through learning and includes a weight of synapse connection and a deviation of a neuron. A hyper-parameter refers to a parameter to be set before learning in a machine learning algorithm and includes a learning rate, the number of iterations, a mini-batch size, and an initialization function.
(315) Learning an artificial neural network may be intended to determine a model parameter for minimizing a loss function. The loss function may be used as an index for determining an optimal model parameter in a process of learning the artificial neural network.
(316) Machine learning may be classified into supervised learning, unsupervised learning, and reinforcement learning.
(317) Supervised learning refers to a method of training an artificial neural network with a label given for training data, wherein the label may indicate a correct answer (or result value) that the artificial neural network needs to infer when the training data is input to the artificial neural network. Unsupervised learning may refer to a method of training an artificial neural network without a label given for training data. Reinforcement learning may refer to a training method for training an agent defined in an environment to choose an action or a sequence of actions to maximize a cumulative reward in each state.
(318) Machine learning implemented with a deep neural network (DNN) including a plurality of hidden layers among artificial neural networks is referred to as deep learning, and deep learning is part of machine learning. Hereinafter, machine learning is construed as including deep learning.
(319) The foregoing technical features may be applied to wireless communication of a robot.
(320) Robots may refer to machinery that automatically process or operate a given task with own

ability thereof. In particular, a robot having a function of recognizing an environment and autonomously making a judgment to perform an operation may be referred to as an intelligent robot.
(321) Robots may be classified into industrial, medical, household, military robots and the like according uses or fields. A robot may include an actuator or a driver including a motor to perform various physical operations, such as moving a robot joint. In addition, a movable robot may include a wheel, a brake, a propeller, and the like in a driver to run on the ground or fly in the air through the driver.
(322) The foregoing technical features may be applied to a device supporting extended reality.
(323) Extended reality collectively refers to virtual reality (VR), augmented reality (AR), and mixed reality (MR). VR technology is a computer graphic technology of providing a real-world object and background only in a CG image, AR technology is a computer graphic technology of providing a virtual CG image on a real object image, and MR technology is a computer graphic technology of providing virtual objects mixed and combined with the real world.
(324) MR technology is similar to AR technology in that a real object and a virtual object are displayed together. However, a virtual object is used as a supplement to a real object in AR technology, whereas a virtual object and a real object are used as equal statuses in MR technology.
(325) XR technology may be applied to a head-mount display (HMD), a head-up display (HUD), a mobile phone, a tablet PC, a laptop computer, a desktop computer, a TV, digital signage, and the like. A device to which XR technology is applied may be referred to as an XR device.
(326) The claims recited in the present specification may be combined in a variety of ways. For example, the technical features of the method claim of the present specification may be combined to be implemented as a device, and the technical features of the device claims of the present specification may be combined to be implemented by a method. In addition, the technical characteristics of the method claim of the present specification and the technical characteristics of the device claim may be combined to be implemented as a device, and the technical characteristics of the method claim of the present specification and the technical characteristics of the device claim may be combined to be implemented by a method.

Claims

1. A method in a wireless local area network (WLAN) system, the method comprising: receiving, by a non-access point (AP) station (STA), a Management frame related to Enhanced Distributed Channel Access (EDCA) parameter information from an AP, wherein the Management frame includes an EDCA Parameter set which is individually addressed; determining, by the non-AP STA, EDCA parameters of the non-AP STA based on the Management frame; and based on the Management frame, transmitting, by the non-AP STA, an Extremely High Throughput (EHT) physical protocol data unit (PPDU) including a legacy signal (L-SIG) field, a repeated L-SIG (RL-SIG) field being contiguous to the L-SIG field, a Universal Signal (U-SIG) field being contiguous to the RL-SIG field, wherein a length field in the L-SIG field has a value satisfying a condition a remainder is zero when the length field is divided by three, wherein the remainder is used to differentiate the EHT PPDU from a High Efficiency (HE) PPDU, wherein four extra subcarriers are inserted at indexes of {−28, −27, 27, 28} in the RL-SIG field, and values of the four extra subcarriers are {−1, −1, −1, 1}, respectively, wherein the U-SIG field has a length of two symbols, wherein each symbol of the U-SIG field carries 26-bit information, wherein the U-SIG field includes version-independent bits followed by version-dependent bits, wherein version-independent bits include 3-bit information related to a Physical (PHY) version, 1-bit formation indicating whether the EHT PPDU is sent in uplink or downlink, Basic Service Set (BSS) Color information related to an identifier of a BSS, and Transmission Opportunity (TXOP) information related to a TXOP duration.

2. The method of claim 1, further comprising: transmitting, by the non-AP STA, an Association Request frame to the AP, the Association Request frame includes first capability information related to whether the non-AP STA supports priority access related to communicating traffic with a higher priority; and receiving, by the non-AP STA, an Association Response from the AP, the Association Response frame includes second capability information related to whether the AP supports the priority access.
3. The method of claim 2, wherein the priority access is supported based on low-latency access.
4. A non-Access Point (AP) station (STA) configured to operate in a wireless local area network (LAN) system, comprising: at least one processor; and at least one computer memory operably connectable to the at least one processor and storing instructions that, based on being executed by the at least one processor, perform operations comprising: receiving a Management frame related to Enhanced Distributed Channel Access (EDCA) parameter information from an AP, wherein the Management frame includes an EDCA Parameter set which is individually addressed; determining EDCA parameters of the non-AP STA based on the Management frame; and based on the Management frame, transmitting an Extremely High Throughput (EHT) physical protocol data unit (PPDU) including a legacy signal (L-SIG) field, a repeated L-SIG (RL-SIG) field being contiguous to the L-SIG field, a Universal Signal (U-SIG) field being contiguous to the RL-SIG field, wherein a length field in the L-SIG field has a value satisfying a condition a remainder is zero when the length field is divided by three, wherein the remainder is used to differentiate the EHT PPDU from a High Efficiency (HE) PPDU, wherein the four extra subcarriers are inserted at indexes of {−28, −27, 27, 28} in the RL-SIG field, and values of the four extra subcarriers are {−1, −1, −1, 1}, respectively, wherein the U-SIG field has a length of two symbols, wherein each symbol of the U-SIG field carries 26-bit information, wherein the U-SIG field includes version-independent bits followed by version-dependent bits, wherein the version-independent bits include 3-bit information related to a Physical (PHY) version, 1-bit formation indicating whether the EHT PPDU is sent in uplink or downlink, Basic Service Set (BSS) Color information related to an identifier of a BSS, and Transmission Opportunity (TXOP) information related to a TXOP duration.
5. The non-AP STA of claim 4, wherein the instructions perform further operations comprising: transmitting an Association Request frame to the AP, the Association Request frame includes first capability information related to whether the non-AP STA supports priority access related to communicating traffic with a higher priority; and receiving an Association Response from the AP, the Association Response frame includes second capability information related to whether the AP supports the priority access.
6. The non-AP STA of claim 4, wherein the priority access is supported based on low-latency access.